

PROOF-AUDITED AND EXPANDED BOUNDS AND REDUCTIONS FOR RS-MCA

ABSTRACT. This expanded proof-audited note separates genuine theorem-level material from the still-missing dense-frontier upper ledger for smooth-domain Reed–Solomon MCA. The unsafe side is reduced to exact integer prefix floors and the simple-pole list-to-MCA conversion. On the safe side the note keeps only statements proved under their displayed hypotheses: broad safe anchors, quotient descent, split-pencil reductions, second-moment identities, unconditional but weak prefix/split-pair ceilings, the exact saturation identity for MCA codeword rays, and the theorem that the shift-pair ledger is downstream from Q once Q is stated as a max-fiber bound. This revision also fixes endpoint and notation issues found in the audit, expands the finite-arithmetic tables, adds line-ray saturation bookkeeping, and prints the remaining proof obligations explicitly. It does not claim the adjacent deployed safe rows: the active quantitative tasks remain a row-sharp multilevel prefix max-fiber theorem and a saturated primitive split-pencil line-ray theorem.

1. THE THRESHOLD PROBLEM

Let $C = \text{RS}[\mathbb{F}, D, k]$ be the Reed–Solomon code obtained by evaluating polynomials of degree $< k$ on a smooth multiplicative or circle-type domain D of size n , and put $\rho = k/n$. In the deployed cases D is defined over a base field $\mathbb{B} \subseteq \mathbb{F}$; write $\beta = \log_2 |\mathbb{B}|$. The quantity of interest is the largest radius δ for which the support-wise mutual correlated agreement error is at most the challenge target:

$$\delta_C^*(\varepsilon^*) = \sup\{\delta < 1 - \rho : \varepsilon_{\text{mca}}(C, \delta) \leq \varepsilon^*\}.$$

All statements below use the closed integer-ball convention: agreement at least a means radius $1 - a/n$, and an adjacent threshold certificate has an unsafe statement at a_0 and a safe statement at $a_0 + 1$.

The entropy–subfield candidate, used below as the certificate target, is the envelope

$$\delta_{\text{env}}(\rho, \beta) = 1 - \rho - g^*(\rho, \beta), \quad g^*(\rho, \beta) = \sup\{g : H_2(\rho + g) \geq \beta g\}.$$

The point of the formula is that the base field $\mathbb{B}$, not a larger challenge field, controls the frontier unless a separate transfer theorem is proved.

Definition 1.1 (support-wise CA and MCA at agreement a). *Let $C \subseteq \mathbb{F}^D$ be linear, $|D| = n$, and $1 \leq a \leq n$. A word $h \in \mathbb{F}^D$ is C -explained on $S \subseteq D$ if some $c \in C$ agrees with h on S . A pair (f_1, f_2) is $C^{\equiv 2}$ -explained on S if some $c_1, c_2 \in C$ agree with f_1, f_2 on S .*

For a pair (f_1, f_2) , a finite slope $\gamma \in \mathbb{F}$ is MCA-bad at agreement a if there is a set $S \subseteq D$, $|S| \geq a$, such that $f_1 + \gamma f_2$ is C -explained on S , but (f_1, f_2) is not $C^{\equiv 2}$ -explained on that same S . It is CA-bad at agreement a if $f_1 + \gamma f_2$ is C -explained on some S with $|S| \geq a$, while (f_1, f_2) is not $C^{\equiv 2}$ -explained on any set $T \subseteq D$ with $|T| \geq a$.

With the uniform slope sampler $\gamma \leftarrow \mathbb{F}$, define

$$\varepsilon_{\text{mca}}(C, 1 - a/n) = \frac{1}{|\mathbb{F}|} \max_{f_1, f_2 \in \mathbb{F}^D} |\{\gamma : \gamma \text{ is MCA-bad at agreement } a\}|,$$

and define $\varepsilon_{\text{ca}}(C, 1 - a/n)$ analogously with CA-bad slopes.

Lemma 1.2 (comparison and monotonicity). *For every linear code $C \subseteq \mathbb{F}^D$ and every agreement threshold a ,*

$$\varepsilon_{\text{ca}}(C, 1 - a/n) \leq \varepsilon_{\text{mca}}(C, 1 - a/n).$$

Moreover the bad-slope sets are monotone in the agreement threshold: if $a' \leq a$, then every MCA-bad slope at agreement a is MCA-bad at agreement a' . Consequently the radius function $\delta \mapsto \varepsilon_{\text{mca}}(C, \delta)$ is nondecreasing under the closed integer-ball convention.

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Proof. If a slope is CA-bad for (f_1, f_2) , then some support S of size at least a explains $f_1 + \gamma f_2$, and no support of size at least a explains the pair. In particular, the same S does not explain the pair, so the slope is MCA-bad. Taking maxima gives the first inequality.

For monotonicity, the same witness support S used at agreement a also satisfies $|S| \geq a'$ when $a' \leq a$. Thus it witnesses badness at the lower agreement threshold. Since increasing the closed radius lowers the required integer agreement, ε_{mca} is nondecreasing in δ . $\square$

Definition 1.3 (staircase numerator). *For an MCA row, let $B_C^{\text{MCA}}(a)$ be the maximum number of MCA-bad line parameters with a witness support of size at least a . For a list row, let $B_C^{\text{list}}(a)$ be the maximum number of codewords agreeing with a received word on at least a positions. When the row track is fixed, write $B(a)$ for the relevant numerator. For target ε^* and challenge denominator Q , write $B^* = \lfloor \varepsilon^* Q \rfloor$. An adjacent certificate at a_0 is the pair of inequalities $B(a_0) > B^*$ and $B(a_0 + 1) \leq B^*$.*

Lemma 1.4 (integer budget convention). *Let a finite sampler have denominator Q , and let B be the number of bad samples. For $B^* = \lfloor \varepsilon^* Q \rfloor$,*

$$\frac{B}{Q} \leq \varepsilon^* \iff B \leq B^*, \quad \frac{B}{Q} > \varepsilon^* \iff B > B^*.$$

Thus every finite adjacent certificate can be stated purely as an integer comparison with B^ .*

Proof. Since B is an integer, $B \leq \varepsilon^* Q$ is equivalent to $B \leq \lfloor \varepsilon^* Q \rfloor$. The strict inequality is the negation of the first condition. $\square$

The unsafe half of the present program supplies the first inequality in Definition 1.3 by identity-prefix floors. The safe half must prove the second inequality after all tangent, quotient, extension, and aperiodic residual cells have been paid without double-counting.

Definition 1.5 (first-match upper ledger). *Fix a row, an agreement a , and a received line (f_1, f_2) . A first-match upper ledger for this line is a finite ordered list of cells $C_1, \dots, C_s$, together with supersets $E_i \subseteq \mathbb{F}$ and certified bounds U_i , such that every MCA-bad slope at agreement a lies in $\bigcup_i E_i$ and $|E_i| \leq U_i$. The associated first-match assignment sends a bad slope γ to the least i with $\gamma \in E_i$.*

Lemma 1.6 (first-match upper ledger). *If every received line admits a first-match upper ledger at agreement a with $\sum_i U_i \leq U$, then $B_C^{\text{MCA}}(a) \leq U$. In particular, $U \leq B^*$ gives the MCA safe half of the adjacent certificate at agreement a .*

Proof. For a fixed line, the first-match sets

$$F_i = \{\gamma : \gamma \text{ is bad and } i \text{ is the least index with } \gamma \in E_i\}$$

are disjoint and cover all bad slopes. Since $F_i \subseteq E_i$, the number of bad slopes is at most $\sum_i |E_i| \leq \sum_i U_i \leq U$. Taking the maximum over received lines gives $B_C^{\text{MCA}}(a) \leq U$. $\square$

Remark 1.7 (MCA versus list rows). The four deployed rows use the same staircase language but not the same lower-side route. The list rows compare the prefix list floor directly with B^* . The MCA rows first form a list floor for $K = k + 1$, then use the deep-point conversion to obtain a bad-slope lower bound. In the moved v13 raw rows this conversion is packaged into an exact integer comparison against B^* , but a certificate must still print which numerator, denominator, and route it uses.

Remark 1.8 (audit convention). Every theorem, proposition, lemma, and corollary below is meant as a proved statement under its displayed hypotheses. A statement is not a proof of the deployed adjacent safe row unless it explicitly supplies the upper inequality $B(a_0 + 1) \leq B^*$ for the row numerator in Definition 1.3. Lower-route stopping statements, quotient-rung vetoes, and raw support censuses are therefore kept separate from upper ledgers. This convention is used to avoid hiding a dense-frontier assumption behind terminology.

2. WHAT IS ALREADY BANKED

The current CAP25 v13 raw manuscript records exact unsafe deployed edges. We treat those as the active experimental source and do not reproduce its long proof here. The table states the consequence needed by this note.

row	target	proved unsafe agreement a_0	unsafe radius
KoalaBear MCA	2^{-128}	1116047	981105/2097152
KoalaBear list	2^{-128}	1116046	490553/1048576
Mersenne-31 MCA	2^{-100}	1116023	981129/2097152
Mersenne-31 list	2^{-100}	1116022	490565/1048576

Each entry is an exact integer comparison of the form

$$\binom{n}{m} > |\mathbb{B}|^w B^*$$

after applying the route specified in Remark 1.7, with the adjacent row failing in the same scanner. The next-step bit margins for the safe inequality are approximately 22.2, 22.0, 3.3, 3.1 bits in the displayed order, so a qualitative $\text{poly}(n)$ loss cannot settle the finite deployed rows.

row	adjacent agreement	spare margin	current adjacent status
KoalaBear MCA	1116048	22.1969 bits	unsafe route stops; no U -certificate
KoalaBear list	1116047	22.0109 bits	unsafe route stops; no U -certificate
Mersenne-31 MCA	1116024	3.2589 bits	tight dyadic watch item, still below budget
Mersenne-31 list	1116023	3.0730 bits	tight dyadic watch item, still below budget

The exact challenge budgets are

$$B_{\text{KB}}^* = \lfloor p_{\text{KB}}^6 / 2^{128} \rfloor = 274980728111395087, \quad B_{\text{M31}}^* = \lfloor p_{\text{M31}}^4 / 2^{100} \rfloor = 16777215.$$

The Mersenne-31 rows therefore use target 2^{-100} , not 2^{-128} ; at 2^{-128} their line field would be below the target denominator and the row would be degenerate. The current packet family records these budgets in

`experimental/data/certificates/frontier-adjacent/*.packet.json`

and the lightweight consistency verifier is

`experimental/scripts/verify_frontier_adjacent_v13_rows.py`.

That verifier checks the stored row constants, the unsafe-side replay, and the full dyadic rung-margin audit. It does not prove $U(a_0 + 1) \leq B^*$, and the JSON packets correctly leave the safe certificate status unset.

Remark 2.1 (finite adjacent artifact status). For the MCA rows, the active lower certificate uses the quantitative deep-list conversion. The packet values are

row	$M(a_0)$	$M(a_0 + 1)$
KoalaBear MCA	138634741058327852652	57198030366
Mersenne-31 MCA	4281388998575706	1752700.

Thus $M(a_0) > B^*$ and the same lower route fails at $a_0 + 1$. This failure is not a safe upper bound. The dyadic rung audit currently has no firing frontier rung at the adjacent agreements; the tightest item is the Mersenne-31 MCA $c = 2048$ quotient watch item, with exact mass $12769758 < 16777215 = B_{\text{M31}}^*$. The extension-cell artifact reduces the full-extension branch to zero-dimensional degree ceilings 4807520, 4226236, 9, 8 for the four rows in table order. These are useful target ceilings, not payments of the corresponding cells.

Remark 2.2 (packet convention). The MCA rows in the current packet family contain historical adjacent-pair fields as well as the active moved-pair block. For v13 raw, the active MCA edges are the moved pairs displayed in this note: KoalaBear MCA at $a_0 = 1116047$ and Mersenne-31 MCA at $a_0 = 1116023$. The list rows are unchanged. In the verifier terminology, the broad dyadic rung scan is gate G6 for the original packet-pair tables, while gate G7 recomputes the

moved MCA lower route and the named Mersenne-31 $c = 2048$ watch item. Thus the rung audit is a veto on currently printed quotient lower-floor refutations at $a_0 + 1$, not a quotient-image upper bound. The Mersenne-31 MCA watch item is close but quiet: it is short by $16777215 - 12769758 = 4007457$ objects.

Theorem 2.3 (quantitative simple-pole list-to-MCA floor). *Let $C = \text{RS}[\mathbb{F}, D, k]$, let $C^+ = \text{RS}[\mathbb{F}, D, k + 1]$, put $q = |\mathbb{F}|$, $n = |D|$, and assume $q > n$. Let $U : D \rightarrow \mathbb{F}$ have $L \geq 1$ distinct codewords of C^+ in the closed Hamming ball of radius δ_0 , and assume $\delta_0 < 1 - k/n$. Then for every $\delta \in [\delta_0, 1 - k/n]$ there is a received line for C with at least*

$$\left\lceil \frac{L(q - n)}{q - n + k(L - 1)} \right\rceil$$

CA-bad finite slopes at radius δ . If $\delta > 0$, the same slopes are support-wise MCA-bad. In particular,

$$\varepsilon_{\text{ca}}(C, \delta), \varepsilon_{\text{mca}}(C, \delta) \geq \frac{1}{q} \left\lceil \frac{L(q - n)}{q - n + k(L - 1)} \right\rceil$$

for $\delta > 0$, with the ε_{mca} conclusion omitted only at the degenerate endpoint $\delta = 0$.

Proof. Let $P_1, \dots, P_L \in \mathbb{F}[X]_{<k+1}$ be distinct polynomials from the δ_0 -ball around U . Since $\delta < 1 - k/n$, every P_i agrees with U on more than k points. For $\alpha \in \Omega := \mathbb{F} \setminus D$, define the simple-pole line

$$f_\alpha(x) = \frac{U(x)}{x - \alpha}, \quad g_\alpha(x) = \frac{-1}{x - \alpha} \quad (x \in D).$$

If P_i agrees with U on S_i , then on S_i

$$f_\alpha(x) + P_i(\alpha)g_\alpha(x) = \frac{P_i(x) - P_i(\alpha)}{x - \alpha},$$

which is the evaluation of a degree- $< k$ polynomial. Thus each distinct value of $P_i(\alpha)$ gives a close slope for the line.

The pair (f_α, g_α) is not column-close at radius δ : a common explanation on more than k points would in particular explain g_α there, so $(X - \alpha)G(X) + 1$ would be a polynomial of degree at most k with more than k roots and value 1 at $X = \alpha$. This is impossible. Therefore all close slopes just constructed are CA-bad, and because the same explaining support cannot explain the pair, they are also MCA-bad whenever the radius is nonzero.

It remains to choose α with many distinct values. For $i \neq j$, the nonzero polynomial $P_i - P_j$ has degree at most k , so $P_i(\alpha) = P_j(\alpha)$ for at most k values $\alpha \in \Omega$. Hence some α has at most $k \binom{L}{2} / (q - n)$ colliding unordered pairs among the L values. If the value multiplicities are $m_1, \dots, m_M$, then

$$\sum_r m_r = L, \quad \sum_r m_r^2 = L + 2 \sum_r \binom{m_r}{2} \leq L + \frac{kL(L - 1)}{q - n}.$$

Cauchy–Schwarz gives

$$L^2 \leq M \sum_r m_r^2, \quad M \geq \frac{L(q - n)}{q - n + k(L - 1)}.$$

Since M is an integer, the displayed ceiling is valid. $\square$

Corollary 2.4 (identity-prefix unsafe criterion). *Let $D \subseteq \mathbb{B} \subseteq \mathbb{F}$, $|D| = n$, and write $b = |\mathbb{B}|$, $q = |\mathbb{F}|$. Fix an agreement a and a target numerator B^* .*

(i) *For the list row $C_K = \text{RS}[\mathbb{F}, D, K]$, put $w = a - K$. If $w \geq 0$ and*

$$\left\lceil \frac{\binom{n}{a}}{b^w} \right\rceil > B^*,$$

then there is a received word with more than B^ codewords of C_K agreeing on at least a positions.*

(ii) For the MCA row $C = \text{RS}[\mathbb{F}, D, k]$, put $K = k + 1$, $w = a - K$, and

$$L_a = \left\lceil \frac{\binom{n}{a}}{b^w} \right\rceil.$$

If $w \geq 0$, $q > n$, and

$$\left\lceil \frac{L_a(q - n)}{q - n + k(L_a - 1)} \right\rceil > B^*,$$

then C has more than B^* support-wise MCA-bad finite slopes at agreement a for some received line.

Proof. For (i), Proposition 3.2 gives a prefix value whose fiber has at least $\lceil \binom{n}{a}/b^w \rceil$ members; those members are distinct degree- $< K$ codewords agreeing with one received word on exactly a positions.

For (ii), apply the same list construction to $C^+ = \text{RS}[\mathbb{F}, D, k + 1]$ and then use Theorem 2.3 at $\delta = 1 - a/n$. Since $a \geq k + 1$, this radius is $< 1 - k/n$, and the displayed ceiling is a lower bound on the number of MCA-bad finite slopes. $\square$

Proposition 2.5 (finite packet consequences). *For the four deployed rows in the table above, let p denote the row's base prime. The current packet family proves the unsafe inequality at the displayed active agreement a_0 . More precisely:*

(i) For the two list rows, the identity-prefix list floor

$$L(a) = \left\lceil \frac{\binom{n}{a}}{p^{a-k}} \right\rceil$$

has the exact values

row	$L(a_0)$	$L(a_0 + 1)$
KoalaBear list	157702518233425975347	65065153468
Mersenne-31 list	4870025984688527	1993678.

Thus $L(a_0) > B^*$ and $L(a_0 + 1) \leq B^*$.

(ii) For the two MCA rows, with $K = k + 1$,

$$L(a) = \left\lceil \frac{\binom{n}{a}}{p^{a-K}} \right\rceil, \quad M(a) = \left\lceil \frac{L(a)(q - n)}{q - n + k(L(a) - 1)} \right\rceil,$$

the active moved-pair packet values are

row	$M(a_0)$	$M(a_0 + 1)$
KoalaBear MCA	138634741058327852652	57198030366
Mersenne-31 MCA	4281388998575706	1752700.

Thus $M(a_0) > B^*$ and $M(a_0 + 1) \leq B^*$.

Consequently the listed a_0 's are genuine unsafe agreements for their rows, but the adjacent inequalities $L(a_0 + 1) \leq B^*$ and $M(a_0 + 1) \leq B^*$ are only lower-route failures, not safe upper bounds.

Proof. For list rows, the displayed integers are obtained by substituting the row values of n, k, p, a_0 into the exact formula for $L(a)$. Comparing them with

$$B_{\text{KB}}^* = 274980728111395087, \quad B_{\text{M31}}^* = 16777215$$

gives $L(a_0) > B^*$ and $L(a_0 + 1) \leq B^*$. By Corollary 2.4(i), the first inequality is exactly the unsafe list certificate; the second says only that this particular prefix-floor construction stops one agreement later.

For MCA rows, Corollary 2.4(ii) says that the resulting bad-slope numerator is at least $M(a)$. Hence $M(a_0) > B^*$ proves $\varepsilon_{\text{mca}}(C, 1 - a_0/n) > \varepsilon^*$. The displayed values are the exact integer values recorded in the active moved-pair packet fields and recomputed by the replay script. Since the same formula gives $M(a_0 + 1) \leq B^*$, the same lower construction no longer proves

failure at the adjacent agreement. This is not an upper bound on all bad slopes, because other tangent, quotient, extension, Q, BC, SP, list, or sparse cells may still contribute. $\square$

Proposition 2.6 (dyadic rung-veto consequence). *Assume the frontier-adjacent packet verifier described above is replayed successfully. Then no currently printed dyadic quotient lower-floor rung among the scanned scales $c = 2^j$, $0 \leq j \leq 20$, and the printed graded-prefix, quotient-remainder, and planted-core profiles refutes any of the four active adjacent candidates $a_0 + 1$. The tightest active frontier watch item is the Mersenne-31 MCA $c = 2048$ rung with exact mass*

$$12769758 < 16777215 = B_{M31}^*.$$

Therefore a new adjacent unsafe certificate must either sharpen a quiet watch item, use an unscanned quotient mechanism, or come from a genuinely aperiodic, extension, sparse/CA, or list/interleaving residual branch.

Proof. This is a scope statement for the exact rung audit, not a new mathematical upper bound. The verifier recomputes, by exact integer arithmetic, every packet field in the dyadic rung tables for the scanned scales and profiles, including whether the corresponding lower floor fires, is within the printed tightness window, and covers the frontier agreement. The stored and recomputed active moved-pair audit has no firing frontier rung at the adjacent agreements; the only active frontier-covering sub-one-bit watch item is the displayed Mersenne-31 MCA $c = 2048$ mass. Since that mass is strictly below B_{M31}^* , the scanned printed quotient lower floors do not themselves refute the adjacent safe candidates. The conclusion is only a veto on this lower-floor family. It does not certify a quotient-image upper ledger $B_Q^{\text{all}}(a_0 + 1)$. $\square$

Proposition 2.7 (unconditional safe-anchor gap at the active adjacent rows). *Let a_+ be the active adjacent agreement $a_0 + 1$ for the four deployed rows, using the moved-pair convention for the two MCA rows. The theorem-level broad safe anchors used in this note begin at agreements*

$$A_{\text{deep}} = 1747627, \quad A_{\text{half}} = 1572864, \quad A_{\text{HJ}} = 1790031.$$

Here A_{deep} is the self-contained deep-MCA edge, A_{half} is the half-distance edge using the BCIKS20 import [BCIKS20], and A_{HJ} is the half-Johnson CA edge. At the active adjacent rows the shortfalls $A - a_+$ are:

row	A_{deep}	A_{half}	A_{HJ}
KoalaBear MCA	631579	456816	673983
KoalaBear list	631580	456817	673984
Mersenne-31 MCA	631603	456840	674007
Mersenne-31 list	631604	456841	674008.

Consequently none of these unconditional broad safe anchors supplies the adjacent upper ledger $U(a_+) \leq B^$. A finite adjacent certificate must use a row-specific first-match upper ledger, or a new theorem that reaches these agreements.*

Proof. The three displayed safe-anchor agreements are the theorem-level thresholds printed in the frontier-adjacent packet family and recomputed by the verifier's safe-theorem applicability audit. The active adjacent agreements are 1116048, 1116047, 1116024, 1116023, respectively; for MCA these are the moved v13 raw adjacent agreements, not the historical packet-pair fields. Subtracting these from the three anchor agreements gives the displayed table. Since each active a_+ is strictly below every listed safe-anchor agreement, the hypotheses of those safe theorems do not reach the active adjacent row. This is only an applicability audit; it gives no upper bound for the residual cells. $\square$

Proposition 2.8 (full-extension cell target). *In the finite-adjacent packet grammar, let $p = |\mathbb{B}|$ and let $a_+ = a_0 + 1$. For a list row set $w = a_0 - k$; for an MCA row set $w = a_0 - k - 1$, because the MCA lower route first builds a list in $\text{RS}[\mathbb{F}, D, k + 1]$. Define the prefix-normalized extension headroom*

$$H_{\text{ext}} := \left\lfloor \frac{p^{w+1} B^*}{\binom{n}{a_+}} \right\rfloor.$$

For the four deployed adjacent rows, in the order used above, exact integer arithmetic gives

row	p	a_+	H_{ext}
KoalaBear MCA	$2^{31} - 2^{24} + 1$	1116048	4807520
KoalaBear list	$2^{31} - 2^{24} + 1$	1116047	4226236
Mersenne-31 MCA	$2^{31} - 1$	1116024	9
Mersenne-31 list	$2^{31} - 1$	1116023	8.

Consequently any full-orbit extension chart paid in this prefix-normalized margin by a dimension-degree term Δp^{e_Y} , with $\Delta \geq 1$, must have $e_Y = 0$, and then must satisfy $\Delta \leq H_{\text{ext}}$ in the corresponding row.

Proof. The displayed H_{ext} is just the exact floor of the row's remaining prefix-normalized multiplicative margin at a_+ . The packet verifier computes it from $n = 2^{21}$, $k = 2^{20}$, the row prime p , the target budget B^* , and the active adjacent agreement a_+ , using integer binomial recurrences rather than floating point. The KoalaBear values use $B^* = \lfloor p^6/2^{128} \rfloor$, while the Mersenne-31 values use $B^* = \lfloor p^4/2^{100} \rfloor$.

If an extension chart contributes Δp^{e_Y} against this margin and $e_Y \geq 1$, then $\Delta p^{e_Y} \geq p$. In all four rows $p > H_{\text{ext}}$: for KoalaBear $p > 2^{30}$ while $H_{\text{ext}} < 2^{23}$, and for Mersenne-31 $p > 2^{30}$ while $H_{\text{ext}} \leq 9$. Thus no positive-dimensional chart can fit the printed adjacent headroom. The zero-dimensional case contributes Δ , so fitting the same margin is exactly the inequality $\Delta \leq H_{\text{ext}}$. $\square$

Remark 2.9 (what the extension target does not prove). The proposition is a target for the unpaid $K = \mathbb{F}$ full-orbit branch, not a payment of that branch. The lower minimal-field branches $K = \mathbb{B}$ and $\mathbb{B} < K < \mathbb{F}$ are routed by confinement and descent to base or lower-arity cells in the v13 raw ledger, while a genuinely full-orbit branch still needs a zero-dimensional eliminant or descended-cycle classification. In particular, the binding Mersenne-31 list row demands total degree at most 8 for the full-orbit residual; a positive-dimensional family would be a new obstruction floor rather than a harmless error term.

The safe side already has two broad anchors. First, a self-contained deep-regime theorem gives MCA error at most $n/|\mathbb{F}|$ below one third of the minimum distance. Second, below one half of the minimum distance, MCA reduces to CA plus a tangent term; with the classical BCIKS half-distance CA input [BCIKS20] this gives the rate-1/2 safe edge $\delta \leq 1/4$. These are not close enough to meet the unsafe edge, but they fix the outside of the remaining band.

Theorem 2.10 (deep-regime MCA bound). *Let $C \subseteq \mathbb{F}^n$ be an $\mathbb{F}$ -linear code with minimum nonzero Hamming weight d , let $q = |\mathbb{F}|$, and put $r = \lfloor \delta n \rfloor$. If*

$$3r \leq d - 1,$$

then

$$\varepsilon_{\text{ca}}(C, \delta) \leq \frac{r+1}{q}, \quad \varepsilon_{\text{mca}}(C, \delta) \leq \frac{r+1}{q}.$$

For $C = \text{RS}[\mathbb{F}, D, k]$, this condition is $3r \leq n - k$.

Proof. Fix a pair (f_1, f_2) , and let

$$S_{\text{cl}} = \{\gamma \in \mathbb{F} : \text{dist}(f_1 + \gamma f_2, C) \leq r\}.$$

If $|S_{\text{cl}}| \leq r + 1$, then this pair contributes at most $r + 1$ close slopes, hence at most $r + 1$ CA- or MCA-bad slopes.

Assume $|S_{\text{cl}}| \geq 2$. Choose distinct $\gamma_1, \gamma_2 \in S_{\text{cl}}$, codewords p_1, p_2 , and error sets E_1, E_2 , $|E_i| \leq r$, such that $f_1 + \gamma_i f_2 = p_i$ off E_i . Define

$$c_2 = \frac{p_1 - p_2}{\gamma_1 - \gamma_2}, \quad c_1 = p_1 - \gamma_1 c_2, \quad e_i = f_i - c_i.$$

Off $T = E_1 \cup E_2$, $|T| \leq 2r$, the two linear equations imply $e_1 = e_2 = 0$. Let $T' \subseteq T$ be the support of the error pair (e_1, e_2) .

For any $\gamma \in S_{\text{cl}}$, with explaining codeword p and error set E , the codeword $p - (c_1 + \gamma c_2)$ vanishes outside $T \cup E$, hence has weight at most $3r \leq d - 1$. It is therefore zero, so

$$\text{wt}(e_1 + \gamma e_2) \leq r \quad (\gamma \in S_{\text{cl}}).$$

If $|T'| \geq r + 1$, each $\gamma \in S_{\text{cl}}$ must make at least $|T'| - r$ of the nonzero affine coordinate functions $e_{1,j} + \gamma e_{2,j}$ vanish. Each coordinate vanishes for at most one γ . Double counting gives

$$|S_{\text{cl}}|(|T'| - r) \leq |T'|, \quad \text{hence} \quad |S_{\text{cl}}| \leq \frac{|T'|}{|T'| - r} \leq r + 1.$$

Thus this pair again contributes at most $r + 1$ close slopes.

It remains to consider $|T'| \leq r$. Then $f_i = c_i$ outside T' , so the pair is column-close to $C^{\equiv 2}$. It has no CA-bad slope. For MCA, if γ has a witness support S of size at least $n - r$, the explaining codeword for $f_1 + \gamma f_2$ agrees with $c_1 + \gamma c_2$ on at least $n - 2r$ positions; since $2r \leq d - 1$, the explaining codeword is $c_1 + \gamma c_2$. A simultaneous explanation of the pair on S would also be forced to be (c_1, c_2) . Hence badness can occur only if $e_1 + \gamma e_2$ vanishes at some coordinate of T' , and each coordinate contributes at most one slope. There are at most $|T'| \leq r$ such slopes. Taking the maximum over pairs proves both bounds. For Reed–Solomon, $d = n - k + 1$. $\square$

Proposition 2.11 (exact high-agreement tangent cell). *Let $C = \text{RS}[\mathbb{F}, D, k]$, $|D| = n$, assume $0 < k < n$, and let $A = n - r$. If $3r \leq n - k$, then the maximum, over received lines, of the number of finite MCA-bad slopes at agreement A is exactly $r + 1 = n - A + 1$. The same exact value holds for CA-bad slopes. Thus the tangent/common-line paid cell is pointwise exact throughout this range.*

Proof. The upper bound is Theorem 2.10 multiplied by $|\mathbb{F}|$.

For the lower bound, choose $T = \{t_0, \dots, t_r\} \subseteq D$ and distinct $\gamma_0, \dots, \gamma_r \in \mathbb{F}$. This is possible because $|\mathbb{F}| \geq n \geq r + 1$. Define

$$f_2(t_i) = 1, \quad f_1(t_i) = -\gamma_i, \quad f_1(x) = f_2(x) = 0 \quad (x \in D \setminus T).$$

For slope γ_i , the word $f_1 + \gamma_i f_2$ vanishes on

$$S_i = D \setminus (T \setminus \{t_i\}), \quad |S_i| = n - r = A,$$

so it is explained there by the zero codeword.

We need two elementary inequalities. If $r = 0$, then $n - r - 1 \geq k$ and $n - 2r - 1 \geq k$. If $r \geq 1$, then $r + 1 \leq 3r \leq n - k$, so $n - r - 1 \geq k$; also $3r \leq n - k$ gives $2r \leq n - k - 1$, hence $n - 2r - 1 \geq k$.

First fix i . If two degree- $< k$ polynomials explained f_1 and f_2 simultaneously on S_i , then both would vanish on $D \setminus T$, a set of size $n - r - 1 \geq k$, hence both would be zero. This contradicts $f_2(t_i) = 1$, since $t_i \in S_i$. Therefore γ_i is MCA-bad at agreement A , and the $r + 1$ slopes are distinct.

For CA-badness, let $S \subseteq D$ have $|S| \geq A$. Since $|T| = r + 1$, the set $S \cap T$ is nonempty. Also

$$|S \cap (D \setminus T)| \geq |S| - |T| \geq n - 2r - 1 \geq k.$$

Any simultaneous degree- $< k$ explanation of f_1, f_2 on S would therefore vanish on at least k points outside T , hence be the zero pair, contradicting $f_2(t) = 1$ for $t \in S \cap T$. Thus the pair is CA-far at agreement A , while each $f_1 + \gamma_i f_2$ is explained on S_i . The same $r + 1$ slopes are CA-bad. $\square$

Remark 2.12 (endpoint audit for the exact tangent cell). The extra hypothesis $k < n$ is only an endpoint issue for the lower construction when $r = 0$. If $k = n$, then $C = \mathbb{F}^D$ and there are no bad slopes, so the exact lower value $r + 1 = 1$ would be false. The upper bound in Theorem 2.10 remains valid for the full-space code.

Theorem 2.13 (MCA from CA below half distance). *Let $C \subseteq \mathbb{F}^n$ be linear with minimum nonzero Hamming weight d , $q = |\mathbb{F}|$, and $r = \lfloor \delta n \rfloor$. If $2r \leq d - 1$, then*

$$\varepsilon_{\text{mca}}(C, \delta) \leq \max\left(\varepsilon_{\text{ca}}(C, \delta), \frac{r}{q}\right).$$

For $C = \text{RS}[\mathbb{F}, D, k]$, the hypothesis is $2r \leq n - k$.

Proof. Fix (f_1, f_2) . If the pair is column-far from $C^{\equiv 2}$ at radius δ , then every MCA-bad slope is already CA-bad: the point $f_1 + \gamma f_2$ is close to C on its witness support, while the pair is far on every support of the required size.

If the pair is column-close, choose codewords $p_1, p_2 \in C$ agreeing with f_1, f_2 outside an error set E , $|E| \leq r$, and write $e_i = f_i - p_i$. Let γ be MCA-bad with witness support S , $|S| \geq n - r$, and explaining codeword c . On $S \setminus E$, the codeword $c - (p_1 + \gamma p_2)$ vanishes; this set has size at least $n - 2r$, so the codeword has weight at most $2r \leq d - 1$ and is zero. Thus on S , the error word $e_1 + \gamma e_2$ vanishes. The support S must meet E , otherwise (p_1, p_2) would explain the pair on S . At some $j \in S \cap E$, we have $e_{1,j} + \gamma e_{2,j} = 0$ with $(e_{1,j}, e_{2,j}) \neq (0, 0)$, forcing $e_{2,j} \neq 0$ and $\gamma = -e_{1,j}/e_{2,j}$. Hence a column-close pair contributes at most $|E| \leq r$ MCA-bad slopes. Taking the maximum over pairs gives the displayed inequality. $\square$

Theorem 2.14 (subfield confinement of bad slopes). *Let $\mathbb{B} \subseteq \mathbb{F}$ be finite fields, let $D \subseteq \mathbb{B}$, and let $C = \text{RS}[\mathbb{F}, D, k]$. If $f, g \in \mathbb{B}^D$, then every support-wise CA- or MCA-bad slope for the line $f + \gamma g$ lies in $\mathbb{B}$. More precisely, if $\gamma \in \mathbb{F} \setminus \mathbb{B}$ and $f + \gamma g$ is explained by a codeword of C on a support $S \subseteq D$, then the pair (f, g) is simultaneously explained by two codewords of C on the same support S . Consequently every $\mathbb{B}$ -valued pair has bad-slope density at most $|\mathbb{B}|/|\mathbb{F}|$ under the uniform $\gamma \leftarrow \mathbb{F}$ sampler. The same conclusion holds after multiplying both words by a common scalar in $\mathbb{F}^\times$.*

Proof. Choose a $\mathbb{B}$ -basis $1 = \beta_0, \beta_1, \dots, \beta_{s-1}$ of $\mathbb{F}$. Every word $h \in \mathbb{F}^D$ decomposes uniquely as $h = \sum_i \beta_i h_i$ with $h_i \in \mathbb{B}^D$. Since $D \subseteq \mathbb{B}$, every codeword of $C = \text{RS}[\mathbb{F}, D, k]$ decomposes componentwise into codewords of $\text{RS}[\mathbb{B}, D, k]$: decompose the coefficients of its defining polynomial in the same basis and evaluate on D .

Let $\gamma = \sum_i \gamma_i \beta_i \notin \mathbb{B}$, and choose $i_* > 0$ with $\gamma_{i_*} \neq 0$. Suppose a codeword $c = \sum_i \beta_i c_i \in C$ agrees with $f + \gamma g$ on S . Comparing basis components on S gives

$$c_0 = f + \gamma_0 g, \quad c_{i_*} = \gamma_{i_*} g.$$

Thus g agrees on S with $\gamma_{i_*}^{-1} c_{i_*} \in C$, and f agrees on S with $c_0 - \gamma_0 \gamma_{i_*}^{-1} c_{i_*} \in C$. Hence every support explaining the line point also explains the pair on the same support.

For MCA this directly rules out badness of every $\gamma \notin \mathbb{B}$. For CA, if $f + \gamma g$ is close on some support of size required by the field radius, the same support gives a column explanation of (f, g) ; hence $\gamma \notin \mathbb{B}$ cannot be CA-bad whenever the internal radius is at least the field radius. Therefore all bad slopes lie in $\mathbb{B}$, giving density at most $|\mathbb{B}|/|\mathbb{F}|$. Multiplication by a common scalar preserves all support explanations by linearity of C , so the same statement holds for scalar multiples of $\mathbb{B}$ -valued pairs. $\square$

3. PROVED PREFIX AND SPLIT-PENCIL REDUCTIONS

We record the part of the Q, BC, and SP program that is already theorem-level. For a finite set $D \subseteq \mathbb{F}$ and $S \subseteq D$, put

$$\ell_S(X) = \prod_{x \in S} (X - x).$$

For $1 \leq w < m \leq n$, define $\Phi_w(M)$ to be the vector of the first w coefficients immediately below the leading term of the monic degree- m polynomial ℓ_M . Thus two m -subsets have the same Φ_w -value exactly when their locator polynomials have the same monic term and same next w high-degree coefficients. Write $\text{Fib}_w(z) = \{M \subseteq D : |M| = m, \Phi_w(M) = z\}$.

Proposition 3.1 (Newton dictionary). *If $\text{char } \mathbb{B} > w$, then Φ_w is triangularly equivalent to the power-sum map*

$$M \mapsto \left(\sum_{x \in M} x, \sum_{x \in M} x^2, \dots, \sum_{x \in M} x^w \right).$$

In particular the two maps have exactly the same fiber sizes.

Proof. Newton's identities express the elementary symmetric functions $e_1, \dots, e_w$ and the power sums $p_1, \dots, p_w$ by mutually inverse triangular polynomial changes of coordinates, because every integer $1, \dots, w$ is invertible in $\mathbb{B}$. The sign convention in Φ_w is coordinatewise invertible, so the fibers are unchanged. $\square$

Proposition 3.2 (identity-prefix witness lists). *Let $K = m - w$. For every prefix value $z = (z_1, \dots, z_w)$, define the received word*

$$U_z(X) = X^m + \sum_{j=1}^w z_j X^{m-j}$$

by evaluation on D . The degree- $< K$ codewords agreeing with U_z on at least m points are exactly

$$U_z - \ell_M, \quad M \in \text{Fib}_w(z).$$

No such codeword agrees with U_z on more than m points. Consequently some z gives a list of size at least $\lceil \binom{n}{m} |\mathbb{B}|^{-w} \rceil$.

Proof. If $M \in \text{Fib}_w(z)$, then the leading $w+1$ coefficients of U_z and ℓ_M agree, so $\deg(U_z - \ell_M) < m - w = K$, and this codeword agrees with U_z on M . Conversely, if a degree- $< K$ polynomial c agrees with U_z on at least m points, then $U_z - c$ is monic of degree m and has those points as roots. It must equal ℓ_M for the corresponding m -set M , and comparison of the first w coefficients puts M in $\text{Fib}_w(z)$. A nonzero degree- m polynomial cannot vanish on more than m points. The average fiber size over the $|\mathbb{B}|^w$ possible prefixes gives the last sentence. $\square$

Proposition 3.3 (pole-line transport to MCA). *Let $C = \text{RS}[\mathbb{F}, D, k]$, fix $k+1 \leq m \leq n$, and put $K = k+1$, $w = m - K$. Assume $z \in \mathbb{B}^w$, $\alpha \in \mathbb{F} \setminus D$, and define*

$$f_\alpha(x) = \frac{U_z(x)}{x - \alpha}, \quad g_\alpha(x) = \frac{-1}{x - \alpha} \quad (x \in D).$$

For a slope $\zeta \in \mathbb{F}$ and a set $S \subseteq D$ with $|S| \geq m$, the word $f_\alpha + \zeta g_\alpha$ is explained by C on S if and only if

$$|S| = m, \quad S \in \text{Fib}_w(z), \quad \ell_S(\alpha) = U_z(\alpha) - \zeta.$$

Moreover the pair (f_α, g_α) is never simultaneously explained by two codewords of C on any set of size at least $k+1$. Hence the finite MCA-bad slopes at agreement threshold m are exactly

$$\{ U_z(\alpha) - \ell_S(\alpha) : S \in \text{Fib}_w(z) \},$$

up to collisions in the displayed evaluation map.

Proof. Suppose $c = P|_D$, $\deg P < k$, explains $f_\alpha + \zeta g_\alpha$ on S . Then $U_z(x) - \zeta = (x - \alpha)P(x)$ on S . The polynomial

$$\Psi(X) = U_z(X) - \zeta - (X - \alpha)P(X)$$

is monic of degree m , because $\deg((X - \alpha)P) \leq k \leq m - 1$, and it vanishes on S . Thus $|S| = m$, $\Psi = \ell_S$, the first w coefficients of ℓ_S are those of U_z , and $\ell_S(\alpha) = U_z(\alpha) - \zeta$. Conversely, if these three conditions hold, then $U_z - \ell_S$ has degree at most k and takes the value ζ at α , so

$$P(X) = \frac{U_z(X) - \ell_S(X) - \zeta}{X - \alpha}$$

has degree $< k$ and explains $f_\alpha + \zeta g_\alpha$ on S .

For the far condition, if g_α agreed on some S , $|S| \geq k+1$, with a polynomial G of degree $< k$, then $(X - \alpha)G(X) + 1$ would have degree at most k and at least $k+1$ roots, hence would be zero. Its value at $X = \alpha$ is 1, a contradiction. A simultaneous explanation of the pair would in particular explain g_α . The final claim follows from the support-wise MCA definition and the first part of the proof. $\square$

Theorem 3.4 (fiber-to-slope conversion). *Let D be a multiplicative coset of order n , let $C = \text{RS}[\mathbb{F}, D, k]$, and let $k+1 \leq m \leq n$, $w = m - k - 1$. Fix a prefix value z and a subfamily*

$\mathcal{G} \subseteq \text{Fib}_w(z)$ of size N_0 . Then there is a pole $\alpha \in \mathbb{F} \setminus D$ for which the line (f_α, g_α) of Proposition 3.3 has at least

$$N_0 - \frac{\binom{N_0}{2}k}{|\mathbb{F}| - n}$$

distinct finite MCA-bad slopes at agreement threshold m . If $N_0 \leq (|\mathbb{F}| - n)/k + 1$, this is at least

$$\frac{N_0}{2} \left(1 - \frac{k}{|\mathbb{F}| - n} \right).$$

If, in addition, $\gcd(m, n) = 1$, every threshold- m witness support of every slope produced by this construction is aperiodic.

Proof. For distinct $S, S' \in \mathcal{G}$, the polynomial $\ell_S - \ell_{S'}$ has degree at most $m - w - 1 = k$, because the leading term and the first w prefix coefficients cancel. Hence the equality $\ell_S(\alpha) = \ell_{S'}(\alpha)$ holds for at most k values of α . Averaging over $\Omega = \mathbb{F} \setminus D$, some $\alpha \in \Omega$ has at most $\binom{N_0}{2}k / (|\mathbb{F}| - n)$ colliding pairs among the values $\ell_S(\alpha)$, $S \in \mathcal{G}$. The image size is at least the number of points minus the number of colliding pairs, giving the first bound. The second bound is the displayed inequality with $N_0 - 1 \leq (|\mathbb{F}| - n)/k$.

By Proposition 3.3, each distinct value $U_z(\alpha) - \ell_S(\alpha)$ is an MCA-bad slope, and every threshold- m witness lies in $\text{Fib}_w(z)$. Finally, if a support $S \subseteq D$ is periodic under a non-trivial subgroup of the order- n stabilizer, then it is a union of orbits of some size $c > 1$ with $c \mid n$, so $c \mid |S| = m$. When $\gcd(m, n) = 1$, this is impossible; all such supports are aperiodic. $\square$

Theorem 3.5 (no uniform polynomial aperiodic bound below the envelope). *There is an infinite family of smooth multiplicative rate-1/2 rows for which, at agreements below the entropy-subfield envelope, a single received line has $2^{\Omega(n)}$ MCA-bad finite slopes, all witnessed aperiodically. Consequently no q -independent polynomial bound can hold for the unnormalized aperiodic band below the envelope.*

Proof. Let $n = 2^\nu$, $k = n/2$, and choose a prime $p_\nu \equiv 1 \pmod{n}$ with $p_\nu = n^{O(1)}$, available by Linnik's theorem [Lin44]. Put $\mathbb{B}_\nu = \mathbb{F}_{p_\nu}$, let $D_\nu = \mu_n \subseteq \mathbb{B}_\nu^\times$, and write $\beta_\nu = \log_2 p_\nu = O(\log n)$. Choose an odd m_ν with

$$\frac{n}{8\beta_\nu} \leq m_\nu - k \leq \frac{n}{4\beta_\nu}$$

and set $w_\nu = m_\nu - k - 1$. Let $\mathbb{F}_\nu/\mathbb{B}_\nu$ be an extension with $q_\nu = |\mathbb{F}_\nu| \geq k 2^{n/8}$, chosen minimally up to the next extension degree.

By Proposition 3.2, some prefix fiber has size at least $\binom{n}{m_\nu} p_\nu^{-w_\nu}$. Since $g = (m_\nu - k)/n \leq 1/(4\beta_\nu)$, the elementary estimate $H_2(1/2 + g) \geq 1 - 4g^2$ gives

$$\log_2 \binom{n}{m_\nu} - w_\nu \beta_\nu \geq n(1 - 4g^2) - ng\beta_\nu - O(\log n) \geq \frac{n}{2}$$

for all large ν . Thus the heaviest fiber has at least $2^{n/2}$ members. By the minimal choice of the extension degree, $\lceil (q_\nu - n)/k \rceil \leq p_\nu 2^{n/8} = 2^{n/8 + O(\log n)}$, so we may choose a subfamily $\mathcal{G}$ of size $N_0 = \lceil (q_\nu - n)/k \rceil$.

Theorem 3.4 gives a pole line with at least

$$\frac{N_0}{2} \left(1 - \frac{k}{q_\nu - n} \right) \geq 2^{n/8-2}$$

bad slopes for all large ν . Because m_ν is odd and n is a power of two, $\gcd(m_\nu, n) = 1$, so every witness support produced by the theorem is aperiodic. This is exponential in n , contradicting any uniform polynomial aperiodic bound in the sub-envelope band. $\square$

Remark 3.6 (what the previous theorem changes). Theorem 3.5 does not weaken the resolution program; it localizes it. The identity-prefix floor is the expected unsafe side below the entropy-subfield envelope, so the positive aperiodic input must be normalized with its left edge at that envelope. Q, BC, and SP are therefore used below as names for safe-side certificate cells after the identity-prefix floor, quotient pullbacks, and planted strata have already been paid.

Proposition 3.7 (prefix-collision rigidity). *If $M \neq M'$ are m -subsets of D with $\Phi_w(M) = \Phi_w(M')$, then*

$$|M \setminus M'| = |M' \setminus M| \geq w + 1.$$

If equality holds, then $\ell_{M \setminus M'} - \ell_{M' \setminus M}$ is a nonzero constant.

Proof. Let $R = M \cap M'$, put $A = \ell_{M \setminus M'}$, $B = \ell_{M' \setminus M}$, and $G = \ell_R$. If $e = |M \setminus M'|$, then

$$\ell_M - \ell_{M'} = G(A - B).$$

The equality of the first w high-degree coefficients gives $\deg(\ell_M - \ell_{M'}) \leq m - w - 1$, while $\deg G = m - e$. Hence $\deg(A - B) \leq e - w - 1$. If $e \leq w$, this forces $A = B$, and then the two disjoint root sets $M \setminus M'$ and $M' \setminus M$ coincide; so $e = 0$, contrary to $M \neq M'$. Thus $e \geq w + 1$. When $e = w + 1$, the same inequality gives $\deg(A - B) \leq 0$; since the root sets are disjoint and nonempty, $A \neq B$, so the constant is nonzero. $\square$

Proposition 3.8 (exact second moment). *Let $N_w(z) = |\text{Fib}_w(z)|$. For $D' \subseteq D$, let $\text{sp}_w(e; D')$ be the number of ordered pairs (A, B) of monic degree- e polynomials split over disjoint root sets in D' such that $\deg(A - B) \leq e - w - 1$. Then*

$$\sum_z N_w(z)^2 = \binom{n}{m} + \sum_{e=w+1}^{\min(m, n-m)} \sum_{\substack{R \subseteq D \\ |R|=m-e}} \text{sp}_w(e; D \setminus R).$$

The top nonzero stratum $e = w + 1$ consists exactly of constant-shift split pairs A and $A - c$, $c \in \mathbb{F}^\times$, both split over D' .

Proof. The left side counts ordered pairs (M, M') of m -subsets with the same prefix. The diagonal contributes $\binom{n}{m}$. For $M \neq M'$, set $R = M \cap M'$, $A = \ell_{M \setminus M'}$, $B = \ell_{M' \setminus M}$, and $e = |M \setminus M'|$. This is a bijective encoding of off-diagonal pairs by R together with a pair (A, B) split over $D \setminus R$ with disjoint root sets. The same degree computation as in Proposition 3.7 shows that prefix equality is precisely $\deg(A - B) \leq e - w - 1$. The range $e \geq w + 1$ follows from Proposition 3.7, and $e \leq \min(m, n - m)$ is forced by the sizes of M and $D \setminus R$. If $e = w + 1$, then $\deg(A - B) \leq 0$, so $A - B = c \in \mathbb{F}^\times$, and the converse is immediate. $\square$

Corollary 3.9 (anticode fiber cap). *For every z , one has*

$$|\text{Fib}_w(z)| \leq \frac{\binom{n}{m-w}}{\binom{m}{w}}.$$

Proof. By Proposition 3.7, two distinct fiber members cannot contain the same $(m - w)$ -subset. Counting pairs (R, M) with $M \in \text{Fib}_w(z)$, $R \subseteq M$, and $|R| = m - w$ gives the claim. $\square$

Proposition 3.10 (moment sandwich). *Let $\mu(z) = |\text{Fib}_w(z)| / \binom{n}{m}$, set*

$$\Gamma_r = |\mathbb{B}|^{w(r-1)} \sum_z \mu(z)^r, \quad r \geq 2,$$

and let $R = |\mathbb{B}|^w \max_z \mu(z)$. Then

$$\frac{\log_2 \Gamma_r}{r-1} \leq \log_2 R \leq \frac{\log_2 \Gamma_r + w \log_2 |\mathbb{B}|}{r}.$$

Proof. Since $\mu(z)^r \leq (\max_z \mu(z))^{r-1} \mu(z)$, we have $\Gamma_r \leq R^{r-1}$. Conversely $\max_z \mu(z) \leq (\sum_z \mu(z)^r)^{1/r}$, and multiplying by $|\mathbb{B}|^w$ gives the upper bound for R . $\square$

Theorem 3.11 (moment criterion for Q). *With the notation of Proposition 3.10, suppose a quotient/planted-pruned prefix problem has normalized maximum-fiber ratio*

$$R = |\mathbb{B}|^w \max_z \frac{|\text{Fib}_w(z)|}{\binom{n}{m}}.$$

Let $r \geq 2$. If

$$\Gamma_r \leq \exp(A) \quad \text{and} \quad w \log |\mathbb{B}| \leq B,$$

then

$$R \leq \exp\left(\frac{A+B}{r}\right).$$

Consequently, an asymptotic Q theorem follows from any sequence of moment bounds with $A = o(rn)$ and $B/r = o(n)$. For a finite adjacent row with remaining bit margin $\Delta > 0$, this moment route can certify $R \leq 2^\Delta$ only if

$$\log_2 \Gamma_r + w \log_2 |\mathbb{B}| \leq r\Delta.$$

Proof. The first bound is the upper half of Proposition 3.10, rewritten with natural logarithms. If $A = o(rn)$ and $B/r = o(n)$, then $\log R = o(n)$, which is the $e^{o(n)}$ form of Q after the paid quotient and planted strata have been removed.

For the finite statement, take logarithms base 2 in the same inequality:

$$\log_2 R \leq \frac{\log_2 \Gamma_r + w \log_2 |\mathbb{B}|}{r}.$$

To prove $R \leq 2^\Delta$, the numerator must be at most $r\Delta$. □

Remark 3.12 (why finite adjacent Q needs high order). At the deployed frontier depths, $w \log_2 |\mathbb{B}|$ is on the order of 10^5 to 10^6 bits, while the adjacent margins are 22.2, 22.0, 3.3, 3.1 bits. Thus a finite adjacent proof using only Theorem 3.11 needs either very high moment order r , an exchange-compression theorem after quotient routing, or a direct max-fiber certificate. Fixed second or third moments can be asymptotically useful evidence, but they cannot by themselves fit the printed adjacent margins. The raw mode-at-null shortcut is false by Proposition 3.14.

Proposition 3.13 (twist-orbit constancy). *Assume D is stable under multiplication by a cyclic group H of order n . For every $\zeta \in H$, the map*

$$(z_1, \dots, z_w) \mapsto (\zeta z_1, \zeta^2 z_2, \dots, \zeta^w z_w)$$

preserves prefix-fiber sizes.

Proof. The map $M \mapsto \zeta M$ is a bijection on m -subsets of D . If

$$\ell_M(X) = X^m + \sum_{j=1}^m \lambda_j(M) X^{m-j},$$

then $\lambda_j(\zeta M) = \zeta^j \lambda_j(M)$. Hence multiplication by ζ sends the fiber over z bijectively to the fiber over $(\zeta^j z_j)_{j \leq w}$. □

Proposition 3.14 (mode-at-null is not the finite Q target). *Let $D = \mathbb{F}_{17}^\times$, $m = 9$, and $w = 1$. For $s \in \mathbb{F}_{17}$, set*

$$N_9(s) = \#\{M \subseteq D : |M| = 9, \sum_{x \in M} x = s\}.$$

Then

$$N_9(0) = 672, \quad N_9(s) = 673 \quad (s \neq 0).$$

Thus the largest first-prefix fiber is not the null fiber. The nonzero fibers form one primitive twist orbit under $\mathbb{F}_{17}^\times$.

Proof. Let $\omega = \exp(2\pi i/17)$. Fourier inversion gives

$$N_9(s) = \frac{1}{17} \sum_{\lambda \in \mathbb{F}_{17}} \omega^{-\lambda s} [T^9] \prod_{x \in \mathbb{F}_{17}^\times} (1 + T\omega^{\lambda x}).$$

For $\lambda = 0$, the coefficient is $\binom{16}{9} = 11440$. For $\lambda \neq 0$, multiplication by λ permutes D , and

$$\prod_{x \in D} (1 + T\omega^{\lambda x}) = \prod_{y=1}^{16} (1 + T\omega^y) = \frac{1 + T^{17}}{1 + T} = 1 - T + T^2 - \dots + T^{16}.$$

The coefficient of T^9 in this last polynomial is -1 . Therefore

$$N_9(0) = \frac{11440 + 16(-1)}{17} = 672, \quad N_9(s) = \frac{11440 - (-1)}{17} = 673 \quad (s \neq 0),$$

because $\sum_{\lambda \neq 0} \omega^{-\lambda s} = -1$ for $s \neq 0$. Since $w = 1$, any nonzero target has twist stabilizer $\gcd(16, 1) = 1$, so the nonzero targets form a full primitive orbit by Proposition 3.13. $\square$

Proposition 3.15 (twist-orbit moment amplification). *Assume D is stable under multiplication by a cyclic group H of order n . For $z = (z_1, \dots, z_w) \in \mathbb{B}^w$, put*

$$A(z) = \{1 \leq j \leq w : z_j \neq 0\}, \quad s(z) = \begin{cases} n, & A(z) = \emptyset, \\ \gcd(n, A(z)), & A(z) \neq \emptyset. \end{cases}$$

Let

$$R(z) = |\mathbb{B}|^w \frac{|\text{Fib}_w(z)|}{\binom{n}{m}}, \quad \Gamma_r = |\mathbb{B}|^{w(r-1)} \sum_y \left(\frac{|\text{Fib}_w(y)|}{\binom{n}{m}} \right)^r \quad (r \geq 2).$$

Then

$$\Gamma_r \geq \frac{n}{s(z)} \frac{R(z)^r}{|\mathbb{B}|^w}.$$

Consequently, if $\Gamma_r \leq G$, then

$$R(z) \leq \left(\frac{G |\mathbb{B}|^w s(z)}{n} \right)^{1/r}.$$

In particular, a primitive prefix value with $\gcd(n, A(z)) = 1$ gains a full factor n in the moment test.

Proof. By Proposition 3.13, every prefix value in the orbit

$$(\zeta z_1, \zeta^2 z_2, \dots, \zeta^w z_w), \quad \zeta \in H,$$

has the same fiber size as z . The stabilizer consists exactly of those $\zeta \in H$ satisfying $\zeta^j = 1$ for every $j \in A(z)$, so its size is $s(z)$. Hence the orbit has size $n/s(z)$. The contribution of this orbit alone to Γ_r is

$$|\mathbb{B}|^{w(r-1)} \frac{n}{s(z)} \left(\frac{|\text{Fib}_w(z)|}{\binom{n}{m}} \right)^r = \frac{n}{s(z)} \frac{R(z)^r}{|\mathbb{B}|^w}.$$

This gives the first inequality, and the second follows by rearranging. $\square$

Proposition 3.16 (composite prefix directions descend). *Let S be a multiplicative coset of order N in $\mathbb{F}_p^\times$, let $e \geq 1$, put $c = \gcd(e, N)$, and let $\psi : \mathbb{F}_p \rightarrow \mathbb{C}^\times$ be a nontrivial additive character. Suppose $g(x) = h(x^e)$. If $S_e = \{a^e : a \in S\}$, then for every m*

$$\sum_{\substack{A \subseteq S \\ |A|=m}} \psi \left(\sum_{a \in A} g(a) \right) = [T^m] \prod_{b \in S_e} (1 + T\psi(h(b)))^c.$$

Thus a composite Fourier direction contributes through the image coset S_e with power-map multiplicity c ; when $c > 1$ it is a quotient-scale object rather than primitive mass.

Proof. The left side is the coefficient of T^m in

$$\prod_{a \in S} (1 + T\psi(g(a))).$$

Since $g(a) = h(a^e)$ is constant on fibers of $a \mapsto a^e$, and the power map has exactly $c = \gcd(e, N)$ preimages over each point of its image on the coset S , the product factors as displayed. $\square$

Theorem 3.17 (proved head-depth prefix flatness). *Let p be prime, $H \leq \mathbb{F}_p^\times$ have order N , put $d = (p-1)/N$, and let $S = \alpha H$. Let $P \subseteq S$ have size ℓ , let $1 \leq w < N - \ell$ satisfy $wd < p$, and put $m' = m - \ell$. For $\vec{z} \in \mathbb{F}_p^w$, define*

$$\mathcal{N}_w^P(m, \vec{z}) = \#\{A : P \subseteq A \subseteq S, |A| = m, (e_1(A), \dots, e_w(A)) = \vec{z}\}.$$

Then

$$\left| \mathcal{N}_w^P(m, \vec{z}) - p^{-w} \binom{N-\ell}{m'} \right| \leq p^{w/2} \binom{w \lceil \sqrt{p} \rceil + \ell + m' - 1}{m'}.$$

For $w = 1$ the prefactor $p^{1/2}$ may be omitted.

Proof. Write $A = P \cup A'$, $A' \subseteq S \setminus P$. Since $\ell_A = \ell_P \ell_{A'}$, the first w elementary symmetric functions of A are an affine unipotent triangular transform of those of A' . Thus planted fibers have the same sizes as unplanted fibers over transformed values, except that the ambient set has size $N - \ell$.

By Fourier inversion over $\mathbb{F}_p^w$, the deviation from the mean is controlled by

$$E(\vec{t}) = \sum_{\substack{A' \subseteq S \setminus P \\ |A'| = m'}} \exp \left(\frac{2\pi i}{p} \sum_{j=1}^w t_j e_j(A') \right), \quad \vec{t} \neq 0.$$

Let j_0 be the largest index with $t_{j_0} \neq 0$. Newton identities rewrite $\sum_j t_j e_j(A')$ as a polynomial in the power sums $p_1(A'), \dots, p_{j_0}(A')$ that is nontrivially linear in the last variable. A second Fourier inversion therefore reduces $E(\vec{t})$ to elementary symmetric functions of the numbers

$$\exp \left(\frac{2\pi i}{p} g(a) \right), \quad a \in S \setminus P,$$

where $g \in \mathbb{F}_p[X]$ is nonconstant of degree at most w .

For every nonzero integer r modulo p , the power sum of this multiset is bounded by $w \lceil \sqrt{p} \rceil + \ell$. Indeed, parameterize the coset as $S = \{\alpha y^d : y \in \mathbb{F}_p^\times\}$. Since $\deg_y(g(\alpha y^d)) \leq wd < p$, Weil's bound for nonconstant additive character sums [Wei48] gives $w\sqrt{p}$ up to harmless endpoint terms; removing the ℓ planted points costs at most ℓ .

Newton's recursion for elementary symmetric functions is coefficientwise dominated by the recursion for $(1 - u)^{-(w \lceil \sqrt{p} \rceil + \ell)}$. Hence every reduced Fourier coefficient is bounded by

$$\binom{w \lceil \sqrt{p} \rceil + \ell + m' - 1}{m'}.$$

The second Fourier transform contributes at most $p^{j_0/2}$ by Parseval, and $j_0 \leq w$. Summing the first Fourier inversion and dividing by p^w gives the stated $p^{w/2}$ bound. When $w = 1$, the Newton-linearization step is absent and the Parseval factor is unnecessary. $\square$

Corollary 3.18 (the solved head of Q). *For the KoalaBear identity-scale subgroup $N = 2^{21}$, $d = 1016$, and $\ell = 0$, Theorem 3.17 proves two-sided prefix flatness at the deployed subset sizes $m = 1116047$ and $m = 1116046$ for exactly $w \leq 21$. At the head rungs $m = K + w$, it is nonvacuous for exactly $w \leq 22$. In these ranges every prefix fiber equals*

$$\binom{N}{m} p^{-w} (1 + \varepsilon)$$

with $|\varepsilon|$ exponentially small in the recorded margin.

Proof. Substitute the displayed row parameters into Theorem 3.17. The nonvacuous range is exactly the range in which the error term is smaller than the main term. Using the standard entropy sandwich

$$\frac{2^{MH_2(r/M)}}{M+1} \leq \binom{M}{r} \leq 2^{MH_2(r/M)}$$

turns this into explicit finite inequalities in p, N, m, w , which are the head-depth inequalities recorded in the v13 raw manuscript. $\square$

Remark 3.19 (why head-depth Q does not close the dense frontier). Theorem 3.17 pays one square-root loss per active power-sum direction. This is why it reaches only $w = 21$ or 22 in the KoalaBear head regimes, while the deployed frontier rows have $w \approx 67466$. The next Q target is therefore not a uniform per-character bound. Composite Fourier directions with nontrivial power-map multiplicity are quotient-scale objects by Proposition 3.16; the $c = 1$ composite directions and primitive character maxima still face the usual Parseval square-root barrier. The remaining statement must bound the maximum prefix fiber itself, after quotient and planted strata have been paid.

Proposition 3.20 (Fourier audit for Q). *Let $D \subseteq \mathbb{F}_p$, let $\Phi_w : \binom{D}{m} \rightarrow \mathbb{F}_p^w$ be the prefix map, and write $N(z) = |\Phi_w^{-1}(z)|$ and $\bar{F} = \binom{|D|}{m} p^{-w}$. For $t \in \mathbb{F}_p^w$, define*

$$E(t) = \sum_{\substack{M \subseteq D \\ |M|=m}} \exp\left(\frac{2\pi i}{p} t \cdot \Phi_w(M)\right).$$

Then

$$N(z) - \bar{F} = p^{-w} \sum_{t \neq 0} \exp\left(-\frac{2\pi i}{p} t \cdot z\right) E(t), \quad \sum_{t \neq 0} |E(t)|^2 = p^w \sum_z |N(z) - \bar{F}|^2.$$

Consequently a proof of Q that uses only the triangle inequality and a uniform nonzero-character estimate must reach the scale

$$\max_{t \neq 0} |E(t)| \leq \text{poly}(n) \binom{|D|}{m} p^{-w}.$$

Uniform square-root estimates of order $\binom{|D|}{m} p^{-w/2}$ are short by a factor $p^{w/2}$ and cannot close the frontier-depth Q input without additional joint cancellation or max-fiber structure.

Proof. The function N on the finite abelian group $\mathbb{F}_p^w$ has Fourier transform E . Fourier inversion gives the first displayed identity, and the zero-frequency term is exactly $\bar{F}$. Plancherel applied to $N - \bar{F}$ gives the second identity, since the zero Fourier coefficient of $N - \bar{F}$ is zero.

If the first formula is bounded by triangle inequality, then

$$|N(z) - \bar{F}| \leq p^{-w} (p^w - 1) \max_{t \neq 0} |E(t)| < \max_{t \neq 0} |E(t)|.$$

Q asks, after quotient and planted strata have been removed, for $N(z) \leq \text{poly}(n) \bar{F}$. Since $\bar{F} = \binom{|D|}{m} p^{-w}$, the required individual-character scale is the one displayed. The square-root scale is larger by $p^{w/2}$. $\square$

Proposition 3.21 (quotient pullback for shift pairs). *Assume D is a multiplicative coset and $c \mid n$. Suppose $A(X) = A_0(X^c)$ and $B(X) = B_0(X^c)$ are a depth- w shift pair of degree $e = ce_0$, meaning $\deg(A - B) \leq e - w - 1$. Then (A_0, B_0) is a shift pair on the quotient coset at depth*

$$w_c = \left\lceil \frac{w+1}{c} \right\rceil - 1.$$

Conversely, any quotient pair at depth w_c pulls back to a depth- w pair upstairs.

Proof. The inequality $c \deg(A_0 - B_0) \leq ce_0 - w - 1$ is equivalent to

$$\deg(A_0 - B_0) \leq e_0 - \left\lceil \frac{w+1}{c} \right\rceil,$$

which is precisely the depth- w_c condition. The converse is the same integer inequality read in the other direction. $\square$

Definition 3.22 (coefficient scale). *Assume $D = \alpha H \subseteq \mathbb{B}^\times$, where H is cyclic of order n and $\text{char } \mathbb{B} \nmid n$. For a monic degree- e polynomial*

$$L(X) = X^e + \sum_{j=1}^e \lambda_j X^{e-j}$$

split over D , define

$$s(L) = \gcd(n, e, \{j : \lambda_j \neq 0\}).$$

For a pair (A, B) of monic split degree- e polynomials, put $s(A, B) = \gcd(s(A), s(B))$.

Lemma 3.23 (coefficient scale detects quotient support). *Let $S \subseteq D$, $|S| = e$, and $L_S = \ell_S$. For $c \mid n$, the following are equivalent:*

$$S \text{ is a union of fibers of } x \mapsto x^c, \quad L_S(X) = L_c(X^c) \text{ for a monic } L_c, \quad c \mid s(L_S).$$

In this case L_c is uniquely determined and split over the quotient coset $D_c = \{x^c : x \in D\}$.

Proof. If S is a union of fibers of $x \mapsto x^c$, then for $\bar{S} = \{x^c : x \in S\}$ one has

$$L_S(X) = \prod_{y \in \bar{S}} (X^c - y) = L_c(X^c).$$

This proves the first implication and uniqueness. The second displayed condition is equivalent to $c \mid e$ and to all nonzero coefficient gaps j being divisible by c , which is exactly $c \mid s(L_S)$. Conversely, if $L_S = L_c(X^c)$, then every root $x \in D$ brings with it the whole fiber of $x \mapsto x^c$, since $X^n - \alpha^n$ is squarefree and $c \mid n$. Thus S is a union of quotient fibers. $\square$

Theorem 3.24 (maximal quotient extraction for shift pairs). *Let (A, B) be a depth- w shift pair of degree e over D :*

$$A = \ell_S, \quad B = \ell_T, \quad S \cap T = \emptyset, \quad \deg(A - B) \leq e - w - 1.$$

Let $c = s(A, B)$. If $c > 1$, then (A, B) is quotient-borne. More precisely, there are unique monic locators A_c, B_c , split over D_c , such that

$$A(X) = A_c(X^c), \quad B(X) = B_c(X^c),$$

and (A_c, B_c) is a depth

$$w_c = \left\lceil \frac{w+1}{c} \right\rceil - 1$$

shift pair on D_c . If c is maximal, then the descended pair has common coefficient scale 1 relative to the quotient length n/c .

Proof. Lemma 3.23 gives the unique quotient locators. The depth statement is exactly Proposition 3.21. For maximality, suppose the descended pair had a nontrivial common coefficient scale $d > 1$ relative to n/c . Then $A_c = A_{cd}(Y^d)$ and $B_c = B_{cd}(Y^d)$, so $A = A_{cd}(X^{cd})$ and $B = B_{cd}(X^{cd})$, contradicting maximality of c . $\square$

Corollary 3.25 (primitive exclusion by coefficient support). *Every primitive shift pair has*

$$\gcd(n, e, \{j : \lambda_j(A) \neq 0\}, \{j : \lambda_j(B) \neq 0\}) = 1.$$

In particular, the top-stratum binomial prototypes $A = X^{w+1} - \beta$, $B = X^{w+1} - \beta'$, when split over D , are quotient-borne and belong to the quotient ledger rather than the primitive SP cell.

Proof. The displayed gcd is $s(A, B)$. If it is > 1 , Theorem 3.24 extracts a quotient pair, so the pair is not primitive. For the binomial prototypes, the only nonzero coefficient gap is $w+1 = e$, and splitting over the order- n coset forces $w+1 \mid n$; hence the common coefficient scale is $w+1$. $\square$

Proposition 3.26 (top-stratum quotient sieve). *Assume $D = \alpha H \subseteq \mathbb{B}^\times$, where H is cyclic of order n and $\text{char } \mathbb{B} \nmid n$. Let $(A, B) = (\ell_S, \ell_T)$ be a depth- w shift pair of degree $e = w+1$ over D , with $S \cap T = \emptyset$, and put $c = s(A, B)$. If $c > 1$, then $c \mid w+1$, and maximal quotient extraction gives unique monic locators A_c, B_c , split over $D_c = \{x^c : x \in D\}$, such that*

$$A(X) = A_c(X^c), \quad B(X) = B_c(X^c),$$

with

$$\deg A_c = \deg B_c = \frac{w+1}{c}, \quad A_c - B_c \in \mathbb{B}^\times.$$

Thus every non-primitive top-stratum shift pair is paid by a top-stratum constant-shift cell on a quotient coset. The primitive top-stratum residual consists only of constant-shift pairs $A, A - \lambda$, $\lambda \in \mathbb{B}^\times$, with $s(A, A - \lambda) = 1$.

Proof. Since $c = s(A, B)$, the definition of coefficient scale gives $c \mid e = w + 1$. By Theorem 3.24, there are unique quotient locators A_c, B_c split over D_c , and the descended pair has depth

$$w_c = \left\lceil \frac{w+1}{c} \right\rceil - 1 = \frac{w+1}{c} - 1.$$

Its degree is $e_c = e/c = (w+1)/c$. Therefore

$$\deg(A_c - B_c) \leq e_c - w_c - 1 = 0.$$

The constant is nonzero: if $A_c = B_c$, then $A = B$, contradicting disjointness of the positive-degree root sets. Hence $A_c - B_c \in \mathbb{B}^\times$.

At the upstairs top stratum $e = w + 1$, the shift-pair condition itself says $\deg(A - B) \leq 0$, so every top-stratum pair is a constant-shift pair. The quotient-borne ones are exactly those with $s(A, B) > 1$, by Theorem 3.24. The remaining top-stratum cell is therefore the coefficient-primitive constant-shift cell $s(A, B) = 1$. $\square$

For later reference, split the off-diagonal sum in Proposition 3.8 into quotient-pulled-back pairs and primitive pairs by assigning every periodic pair to its maximal common quotient scale. Put

$$\bar{F} = \binom{n}{m} |\mathbb{B}|^{-w}, \quad \Gamma_2 = |\mathbb{B}|^w \sum_z \left(\frac{|\text{Fib}_w(z)|}{\binom{n}{m}} \right)^2.$$

Proposition 3.27 (exact primitive L^2 ledger). *With the first-match quotient assignment above,*

$$\Gamma_2 = \frac{1}{\bar{F}} + \frac{\sum_e P_e^{\text{quot}}}{\binom{n}{m} \bar{F}} + \frac{\sum_e P_e^{\text{prim}}}{\binom{n}{m} \bar{F}},$$

where P_e^{quot} and P_e^{prim} are the quotient and primitive parts of the e -th off-diagonal term in Proposition 3.8. Thus SP is exactly the task of bounding the last displayed summand after quotient-pulled-back pairs have been paid.

Proof. Divide the identity in Proposition 3.8 by $\binom{n}{m}^2 |\mathbb{B}|^{-w}$. The diagonal contributes $1/\bar{F}$; the off-diagonal terms split by the first-match quotient/primitive assignment. $\square$

Proposition 3.28 (slope elimination). *Fix K and $m \geq K$. Let $u + zv$ be an affine line of received words and let $T \subseteq D$ have $|T| = m$. Put $w = m - K$ and*

$$S_r(h, T) = \sum_{x \in T} \frac{h(x)x^r}{\ell'_T(x)}, \quad 0 \leq r < w.$$

Then $u + zv$ restricts on T to a polynomial of degree $< K$ if and only if

$$(S_r(u, T))_{r < w} + z(S_r(v, T))_{r < w} = 0.$$

If T is not a common support for both u and v , then it supports at most one finite slope z .

Proof. The Lagrange interpolant of a word $h : T \rightarrow \mathbb{F}$ has degree $< K$ if and only if its top coefficients vanish; the displayed S_r 's are the usual residue formula for those top coefficients, up to an invertible triangular change of basis. Applying this to $h = u + zv$ gives the vector equation. If the second vector is zero, the support is either common or carries no finite slope. If it is nonzero, any solution z is forced by a nonzero coordinate, so there is at most one. $\square$

Proposition 3.29 (interpolation-lattice split-pencil reduction). *Fix K and an agreement $m \geq K$, and write $|D| = n$. For a word $U : D \rightarrow \mathbb{F}$, define*

$$M_U = \{(W, N) \in \mathbb{F}[X]^2 : W(x)U(x) = N(x) \text{ for all } x \in D\},$$

$$\text{wdeg}(W, N) = \max(\deg W, \deg N - (K - 1)).$$

Let $\omega = n - m$. Agreement supports of size m for $C = \text{RS}[\mathbb{F}, D, K]$ are in bijection with pairs $(W, N) \in M_U$ such that W is monic, D -split, $\deg W = \omega$, $\text{wdeg}(W, N) \leq \omega$, and $W \mid N$. If a shifted weak-Popov basis $g_i = (W_i, N_i)$ of M_U has profile (d_1, d_2) , every such pair has the form $Ag_1 + Bg_2$ with $\deg A \leq \omega - d_1$ and $\deg B \leq \omega - d_2$. When $W = AW_1 + BW_2$ is D -split, the divisibility $W \mid N$ is automatic.

Proof. If U agrees with a degree- $< K$ polynomial c on T , take $W = \ell_{D \setminus T}$ and $N = Wc$. Conversely, a pair with $W \mid N$ gives $N = Wc$, the weighted-degree bound gives $\deg c < K$, and the relation $W(U - c) = 0$ on D gives agreement on $D \setminus Z(W)$. The weak-Popov representation is the predictable-degree property of a shifted basis of the rank-two interpolation module. Finally, if W is D -split, then at every root $x \in D$ of W the module relation gives $N(x) = W(x)U(x) = 0$; since W is squarefree, $W \mid N$. $\square$

Theorem 3.30 (near-rational lattice dichotomy). *Let $C = \text{RS}[\mathbb{F}, D, K]$, let $U : D \rightarrow \mathbb{F}$, let $m \geq K$, put $w = m - K$ and $\omega = n - m$, and assume $2w \leq n - K$. Let $d_1 = d_1(U)$ be the smallest shifted degree of a nonzero element of M_U , and let*

$$\text{Cen}(U; m) = \#\{T \subseteq D : |T| = m, U|_T \text{ is explained by a degree-} < K \text{ polynomial}\}.$$

Then exactly one of the following alternatives holds.

- (i) *If $d_1 \leq w$, then either U is within Hamming distance exactly d_1 of a unique codeword $c \in C$, in which case*

$$\text{Cen}(U; m) = \binom{n - d_1}{m},$$

$$\text{or } \text{Cen}(U; m) = 0.$$

- (ii) *If $d_1 \geq w + 1$, every census element is represented in the two-generator family*

$$Ag_1 + Bg_2, \quad \deg A \leq \omega - d_1, \quad \deg B \leq \omega - d_2,$$

where g_1, g_2 is a shifted weak-Popov basis of M_U , $d_2 = n - K + 1 - d_1$. The parameter space has dimension $\omega - w + 1$.

Proof. Let $g_1 = (W_1, N_1)$, $g_2 = (W_2, N_2)$ be a shifted weak-Popov basis with row degrees $d_1 \leq d_2$. The determinant degree gives $d_1 + d_2 = n - K + 1$. If $d_1 \leq w$, then $d_2 \geq n - K + 1 - w = \omega + 1$. By predictable degrees, any element of shifted degree at most ω must be a multiple Ag_1 . Therefore the census is empty unless W_1 , after scaling, is a monic squarefree divisor of ℓ_D , $W_1 \mid N_1$, and $c = N_1/W_1$ has degree $< K$. In the nonempty case, c agrees with U outside the roots of W_1 . Minimality of d_1 forces the disagreement set to be exactly those roots; otherwise its error-locator pair would be a nonzero lattice element of smaller shifted degree. The codeword is unique because two codewords within distance w of U would differ in at most $2w \leq n - K$ positions, contradicting the Reed–Solomon distance $n - K + 1$. The valid supports are precisely the m -subsets of the $n - d_1$ agreement positions, giving the binomial count. If any required split/divisibility/codeword condition fails, the census is empty.

If $d_1 \geq w + 1$, both weak-Popov rows may occur. Predictable degrees give the multiplier bounds $\deg A \leq \omega - d_1$ and $\deg B \leq \omega - d_2$. The number of coefficients in A and B is

$$(\omega - d_1 + 1) + (\omega - d_2 + 1) = 2\omega + 2 - (n - K + 1) = \omega - w + 1.$$

$\square$

Corollary 3.31 (near-rational slopes are paid outside the balanced core). *For $C = \text{RS}[\mathbb{F}, D, K]$ and an affine line $u + zv$, suppose two distinct slopes z_1, z_2 have $d_1(u + z_i v) \leq w$ and nonzero size- m census. Then v is within distance $2w$ of a codeword and u is within distance $3w$ of a codeword. Thus this line belongs to a common-proximity paid branch; after that branch is removed, at most one near-rational finite slope remains, and all other residual slopes satisfy $d_1(u + zv) \geq w + 1$.*

Proof. By Theorem 3.30, $u + z_i v = c_i + \eta_i$, with $c_i \in C$ and $\text{wt}(\eta_i) \leq w$, for $i = 1, 2$. Since $z_1 \neq z_2$,

$$v = \frac{c_1 - c_2}{z_1 - z_2} + \frac{\eta_1 - \eta_2}{z_1 - z_2}$$

is within distance at most $2w$ of a codeword. Substituting into $u = (u + z_1v) - z_1v$, the word u is within distance at most $w + 2w = 3w$ of a codeword. Such a line is therefore not part of the primitive balanced-core residual once common-proximity/tangent cells are paid. If two near-rational slopes survived there, the preceding argument would put the line in the paid branch; hence at most one can remain. $\square$

Theorem 3.32 (deficiency-one split-test eliminant). *Let $\Lambda_D \in \mathbb{F}[X]$ be squarefree of degree n . Let $M(Z) = M_0 + ZM_1$ be a $j \times (j+1)$ matrix pencil over $\mathbb{F}$. For $0 \leq i \leq j$, let $c_i(Z)$ be the signed maximal minor obtained by deleting column i , and set*

$$L_Z(X) = \sum_{i=0}^j c_i(Z)X^i.$$

Assume not all maximal minors vanish identically. If $c_j \neq 0$, perform pseudo-division over $\mathbb{F}[Z]$:

$$c_j(Z)^{n-j+1}\Lambda_D(X) = Q(Z, X)L_Z(X) + R(Z, X), \quad \deg_X R < j.$$

If some coefficient $R_m(Z)$ is not identically zero, then every parameter z for which either $M(z)$ drops rank, or $M(z)$ has full rank with $c_j(z) = 0$, or $M(z)$ has full rank with $c_j(z) \neq 0$ and $L_z \mid \Lambda_D$, is a root of the nonzero polynomial $c_j(Z)R_m(Z)$. Its degree is at most $j + (n - j + 1)j$. If $c_j \equiv 0$, the top chart is empty. If $c_j \neq 0$ and all coefficients of R vanish identically, then every top-chart parameter passes the split test $L_z \mid \Lambda_D$, so this chart is a named residual branch rather than a paid bounded chart.

Proof. Each maximal minor is a determinant of a $j \times j$ matrix with affine-linear entries in Z , so it has degree at most j . At a full-rank value z , the cofactor vector $c(z) = (c_0(z), \dots, c_j(z))$ spans the one-dimensional kernel of $M(z)$. Thus $c_j(z) = 0$ is the full-rank chart where the candidate locator has degree $< j$, while $c_j(z) \neq 0$ is the top chart.

Pseudo-division by the degree- j polynomial L_Z takes $n - j + 1$ elimination steps. Each step multiplies the current remainder by c_j and subtracts a shifted multiple of L_Z , so every coefficient of R has Z -degree at most $(n - j + 1)j$.

Let z be a top-chart parameter with $L_z \mid \Lambda_D$. Since $c_j(z) \neq 0$, substituting $Z = z$ into the pseudo-division identity gives that L_z divides $R(z, X)$. But $\deg_X R(z, X) < j = \deg L_z$, hence $R(z, X) = 0$; in particular $R_m(z) = 0$. If $M(z)$ drops rank, all maximal minors vanish, including $c_j(z)$. If $M(z)$ has full rank but $c_j(z) = 0$, again $c_j(z) = 0$. These two classes are therefore roots of $c_j R_m$. The degree bound follows from the preceding paragraph and $\deg c_j \leq j$. Finally, if all coefficients of R vanish identically, then the same pseudo-division identity gives $L_z \mid \Lambda_D$ for every top-chart z , and no eliminant is available. $\square$

Lemma 3.33 (split annihilator dictionary). *Let $D \subseteq \mathbb{F}$ have size n , let $\Lambda_D(X) = \prod_{x \in D} (X - x)$, and put $v_x = \Lambda'_D(x)^{-1}$. Let $L \in \mathbb{F}[X]$ be split over D , with root set T , $\gamma = |T| = \deg L$, and let $\tau \geq 1$ satisfy $\tau + \gamma \leq n$. For a word $y : D \rightarrow \mathbb{F}$, the conditions*

$$\sum_{x \in D} v_x y(x) L(x) x^m = 0 \quad (0 \leq m < \tau)$$

hold if and only if

$$y \in \text{RS}[\mathbb{F}, D, n - \tau - \gamma] + \mathbb{F}^T,$$

where $\mathbb{F}^T$ denotes words supported on T . The sum is direct.

Proof. First note the residue identity: for every polynomial h with $\deg h \leq n - 2$,

$$\sum_{x \in D} v_x h(x) = 0.$$

Indeed, h/Λ_D has no residue at infinity, and its finite residues are $v_x h(x)$.

If y is supported on T , then $y(x)L(x) = 0$ for all $x \in D$. If $y = p|_D$ with $\deg p < n - \tau - \gamma$, then for $m < \tau$ the polynomial pLX^m has degree at most $n - 2$, so the residue identity gives the displayed annihilator equations. Thus the right side lies in the solution space.

Conversely, the linear map

$$y \mapsto \left(\sum_{x \in D} v_x y(x) L(x) x^m \right)_{0 \leq m < \tau}$$

has zero columns on T . On $D \setminus T$, the nonzero scalars $v_x L(x)$ multiply a $\tau \times (n - \gamma)$ Vandermonde matrix on distinct points, hence the rank is τ . The solution space therefore has dimension $n - \tau$. The right side has dimension $(n - \tau - \gamma) + \gamma = n - \tau$, because a degree- $< n - \tau - \gamma$ polynomial vanishing on $D \setminus T$ has at least $n - \gamma$ roots, more than its degree unless it is zero. The containment of equal-dimensional spaces proves the equivalence and directness. $\square$

Lemma 3.34 (constant divisor rigidity). *Let $P \in \mathbb{F}[X]$ be nonzero and let $L \in \mathbb{F}(Z)[X]$ be monic. If $L \mid P$ in $\mathbb{F}(Z)[X]$, then $L \in \mathbb{F}[X]$ and $L \mid P$ already in $\mathbb{F}[X]$.*

Proof. Over $\overline{\mathbb{F}}(Z)$, all roots of P are constants. Since L is a monic divisor of P , it is a product of some of the linear factors $X - \xi$ with $\xi \in \overline{\mathbb{F}}$. Thus the coefficients of L lie in $\overline{\mathbb{F}}$. They also lie in $\mathbb{F}(Z)$, and the relative algebraic closure of $\mathbb{F}$ in $\mathbb{F}(Z)$ is $\mathbb{F}$; hence the coefficients lie in $\mathbb{F}$. Division with remainder in $\mathbb{F}[X]$ agrees with division in $\mathbb{F}(Z)[X]$, so the remainder is zero. $\square$

Proposition 3.35 (split charts are tangent-borne). *Let $C = \text{RS}[\mathbb{F}, D, k]$, set $j = n - a$, and assume $j = a - k$. Let $M(Z)$ be the deficiency-one syndrome pencil of a received line $f + Zg$. In the notation of Theorem 3.32, assume $c_j \neq 0$ and all pseudo-division remainders vanish identically. Then there is a fixed j -set $T^* \subseteq D$ such that f and g each agree with codewords of C on $D \setminus T^*$. Consequently, at every threshold $a' \geq a$, the line has at most j MCA-bad finite slopes, all among the tangent ratios*

$$-\frac{e_f(x)}{e_g(x)}, \quad x \in T^*, \quad e_g(x) \neq 0,$$

where e_f, e_g are the two error words supported on T^* . It has no CA-bad slopes whenever the internal radius is at least j/n .

Proof. The vanishing of all pseudo-division remainders says that the normalized top-chart locator

$$\widehat{L}_Z(X) = c_j(Z)^{-1} \sum_{i=0}^j c_i(Z) X^i$$

divides Λ_D in $\mathbb{F}(Z)[X]$. By Lemma 3.34, $\widehat{L}_Z = L^* \in \mathbb{F}[X]$ is a fixed monic divisor of Λ_D . Let T^* be its root set in D , so $|T^*| = j$.

Because the cofactor vector $c(Z)$ is a kernel vector of the syndrome pencil, the identity $M(Z)c(Z) \equiv 0$ separates into the two identities saying that L^* annihilates the exact-agreement syndromes of f and of g . Applying Lemma 3.33 with $\tau = j$ and $\gamma = j$ gives $f = c_f + e_f$ and $g = c_g + e_g$, with $c_f, c_g \in \text{RS}[\mathbb{F}, D, n - 2j] = C$ and e_f, e_g supported on T^* . Here $n - 2j = k$ by $j = n - a = a - k$.

Let z be MCA-bad at agreement $a' \geq a$, with witness support S and explaining codeword c for $f + zg$. On $S \setminus T^*$ the codewords c and $c_f + zc_g$ agree. This set has size at least $a' - j \geq a - j = k$, so they are the same codeword. Thus on $S \cap T^*$ one has $e_f + ze_g = 0$. If the pair were simultaneously explained on S , it would be explained by (c_f, c_g) , so badness forces some $x \in S \cap T^*$ with $(e_f(x), e_g(x)) \neq (0, 0)$. At such an x , the equality $e_f(x) + ze_g(x) = 0$ forces $e_g(x) \neq 0$ and $z = -e_f(x)/e_g(x)$. There are at most $|T^*| = j$ such ratios. Finally, if the internal radius is at least j/n , then (f, g) is column-close to (c_f, c_g) and therefore cannot be CA-bad. $\square$

Proposition 3.36 (the boundary profile is Q). *Assume D is a multiplicative coset with domain polynomial $X^n - \beta$. Let*

$$P_z(X) = X^m + \sum_{j=1}^w z_j X^{m-j}, \quad X^n = P_z Q_z + R_z, \quad \deg R_z < m,$$

and put $\omega = n - m$. In the interpolation-lattice boundary branch of shifted degree $w + 1$, the prefix fiber is exactly the prescribed-divisor census

$$|\text{Fib}_w(z)| = \#\{A : \deg A \leq \omega - w - 1, Q_z + A \mid X^n - \beta\}.$$

Consequently Q is the boundary profile of the split-pencil problem, and BC starts only after this profile is removed.

Proof. The pair $(1, P_z)$ is in the interpolation module of U_z . The Euclidean division above gives the second vector $(Q_z, \beta - R_z)$, since for every $x \in D$ one has $Q_z(x)P_z(x) = \beta - R_z(x)$. Their determinant is $\beta - X^n$, so they form a basis. In the boundary branch, monic elements of W -degree ω are exactly $Q_z + A$ with $\deg A \leq \omega - w - 1$. Being a valid locator is precisely the divisibility condition $Q_z + A \mid X^n - \beta$. This is the same set counted by $\text{Fib}_w(z)$, by Proposition 3.2. $\square$

Theorem 3.37 (interior base-field split-pencil floor). *Let $C = \text{RS}[\mathbb{F}, D, K]$, where $D \subseteq \mathbb{B} \subseteq \mathbb{F}$, $|D| = n$, and put $b = |\mathbb{B}|$. Fix an agreement $m \geq K$, put $w = m - K$, and let*

$$w + 2 \leq d_1 \leq \left\lfloor \frac{n - K + 1}{2} \right\rfloor.$$

Set $m_{d_1} = K - 1 + d_1$, and assume $m \leq m_{d_1} \leq n$ and $m_{d_1} + d_1 \leq n$. Then there is a $\mathbb{B}$ -valued received word U whose interpolation module has minimal shifted degree exactly d_1 , and whose size- m split-pencil census is at least

$$\binom{m_{d_1}}{m} \left[\binom{n}{m_{d_1}} b^{-(d_1-1)} \right].$$

Consequently any BC upper model for genuine interior profiles must include a base-field-normalized floor; a challenge-field-only normalization is false when $|\mathbb{F}|$ is much larger than $|\mathbb{B}|$.

Proof. For $z = (z_1, \dots, z_{d_1-1}) \in \mathbb{B}^{d_1-1}$, let

$$U_z(X) = X^{m_{d_1}} + \sum_{h=1}^{d_1-1} z_h X^{m_{d_1}-h}$$

on D . The module element $(1, U_z)$ has shifted degree $m_{d_1} - (K - 1) = d_1$. If $(W, N) \in M_{U_z}$ had shifted degree $< d_1$, then $\deg W < d_1$, $\deg N < m_{d_1}$, and $WU_z - N$ would vanish on D . Its degree is at most $m_{d_1} + d_1 - 1 < n$, so $WU_z = N$ as polynomials. If $W \neq 0$, monicity of U_z gives $\deg(WU_z) \geq m_{d_1}$, contradicting $\deg N < m_{d_1}$; hence $W = N = 0$. Thus the minimal shifted degree is exactly d_1 .

Now vary z . The first $d_1 - 1$ high-degree coefficients of the locators of m_{d_1} -subsets of D lie in $\mathbb{B}^{d_1-1}$, so by pigeonhole some z^* has at least

$$\left[\binom{n}{m_{d_1}} b^{-(d_1-1)} \right]$$

subsets M' with that prefix. For each such M' , the polynomial $U_{z^*} - \ell_{M'}$ has degree $< K$, because the leading term and the first $d_1 - 1$ coefficients cancel and $m_{d_1} - d_1 = K - 1$. Hence it agrees with U_{z^*} on every point of M' .

Every m -subset $T \subseteq M'$ is therefore an agreement support. Distinct pairs (M', T) give distinct codeword-support pairs: if the same degree- $< K$ codeword agreed on the same T with $|T| = m \geq K$, then the codewords coincide, and then $\ell_{M'}$ coincides as well. By Proposition 3.29, these agreement-support pairs are represented in the split-pencil count of any shifted weak-Popov basis of $M_{U_{z^*}}$. Since the minimal shifted degree is d_1 , the family lies in the genuine interior profile d_1 , not in the boundary- Q profile. $\square$

Proposition 3.38 (rank-one pole-line base-field floor). *Let $C = \text{RS}[\mathbb{F}, D, K]$, with $D \subseteq \mathbb{B} \subsetneq \mathbb{F}$, and fix $m \geq K + 1$, $w = m - K$. Let $z^* \in \mathbb{B}^w$ be a heaviest depth- w prefix value among m -subsets*

of D , and choose $\alpha \in \mathbb{F} \setminus (D \cup \mathbb{B})$. For the pole line

$$f_\alpha(x) = \frac{U_{z^*}(x)}{x - \alpha}, \quad g_\alpha(x) = \frac{-1}{x - \alpha},$$

there are at least

$$\left\lceil \binom{n}{m} |\mathbb{B}|^{-w} \right\rceil$$

non-common rank-one agreement supports of size m . Moreover the line is genuinely extension-valued, even after common scalar multiplication.

Proof. By pigeonhole, $|\text{Fib}_w(z^*)| \geq \lceil \binom{n}{m} |\mathbb{B}|^{-w} \rceil$. For each $S \in \text{Fib}_w(z^*)$, Proposition 3.3 supplies the slope $\zeta_S = U_{z^*}(\alpha) - \ell_S(\alpha)$ for which $f_\alpha + \zeta_S g_\alpha$ is explained on S . In the slope-elimination notation of Proposition 3.28, this is a rank-one relation

$$(S_r(f_\alpha, S))_{r < w} + \zeta_S (S_r(g_\alpha, S))_{r < w} = 0.$$

It is non-common because g_α cannot be explained on any set of size at least $K + 1$, by the same argument as in Proposition 3.3: otherwise $(X - \alpha)G + 1$ would have degree at most K , vanish on at least $K + 1$ points, and take value 1 at α .

Finally, suppose some nonzero scalar λ made $\lambda g_\alpha(D) \subseteq \mathbb{B}$. For distinct $x, y \in D$, the ratio $g_\alpha(x)/g_\alpha(y) = (y - \alpha)/(x - \alpha)$ would lie in $\mathbb{B}$. Writing this ratio as $r \neq 1$ gives $\alpha = (y - rx)/(1 - r) \in \mathbb{B}$, a contradiction. $\square$

Proposition 3.39 (extension-valued distinct-slope rank-one floor). *Let $C = \text{RS}[\mathbb{F}, D, K]$, with $D \subseteq \mathbb{B} \subsetneq \mathbb{F}$, put $q = |\mathbb{F}|$, and fix $m \geq K + 1$, $w = m - K$. Let $z^* \in \mathbb{B}^w$ be a prefix value and let $\mathcal{G} \subseteq \text{Fib}_w(z^*)$ have size N . Then there exists $\alpha \in \mathbb{F} \setminus \mathbb{B}$ such that the pole line*

$$f_\alpha(x) = \frac{U_{z^*}(x)}{x - \alpha}, \quad g_\alpha(x) = \frac{-1}{x - \alpha}$$

has at least

$$N - \frac{\binom{N}{2}(K - 1)}{q - |\mathbb{B}|}$$

distinct finite MCA-bad slopes at agreement m . In particular, if $N \leq (q - |\mathbb{B}|)/(K - 1) + 1$, then some genuinely extension-valued pole line has at least $N/2$ distinct MCA-bad slopes. The same conclusion applies to any $N \leq \lceil \binom{n}{m} |\mathbb{B}|^{-w} \rceil$ by choosing $\mathcal{G}$ inside a heaviest prefix fiber.

Proof. For each $S \in \mathcal{G}$, Proposition 3.3 gives the bad slope

$$\zeta_S(\alpha) = U_{z^*}(\alpha) - \ell_S(\alpha).$$

If $S \neq S'$, then $\ell_S - \ell_{S'}$ has degree at most $m - w - 1 = K - 1$, because the leading term and the first w high-degree coefficients cancel. Hence $\zeta_S(\alpha) = \zeta_{S'}(\alpha)$ for at most $K - 1$ values of $\alpha \in \mathbb{F}$, and therefore for at most $K - 1$ values in $\mathbb{F} \setminus \mathbb{B}$.

Averaging over the $q - |\mathbb{B}|$ allowed extension poles gives some $\alpha \notin \mathbb{B}$ with at most $\binom{N}{2}(K - 1)/(q - |\mathbb{B}|)$ colliding support-pairs. The number of distinct slope values is at least N minus that number of colliding pairs. By Proposition 3.3, all these slopes are MCA-bad at agreement m , because the pole pair is never simultaneously explained on any support of size at least $K + 1$. Finally, since $\alpha \notin \mathbb{B}$, the scalar-confinement argument from Proposition 3.38 shows that the line is genuinely extension-valued. $\square$

Remark 3.40 (proved reductions versus dense flatness). The propositions and theorems in this section, together with the safe anchors above, are proved reductions, solved subcases, or route-cut counterexamples: prefix exactness, pole-line transport, fiber-to-slope conversion, exact high-agreement tangent cells, subfield confinement, the exponential aperiodic floor below the envelope, collision rigidity, second moments, twist-orbit moment amplification, quotient descent, coefficient-scale quotient extraction, the top-stratum quotient sieve, head-depth flatness, slope elimination, near-rational lattice classification, deficiency-one eliminants, identically split chart collapse to tangent cells, split pencils, the $\mathbb{Q}$ boundary profile, rank-one pole-line floors with

distinct extension-valued slope realization, and base-field census floors. They do not by themselves prove prefix flatness at the deployed frontier depth, the interior split-pencil census, or the finite adjacent safe inequalities. They replace those questions by explicit fiber, slope, pencil, quotient, extension, and primitive shift-pair counts, and they prove the head-depth part of Q outright.

4. PROPER THEOREMS FOR THE FORMER Q/BC/SP CELLS

The labels Q, BC, and SP are retained only as names for residual safe-side mechanisms. The section records actual theorems rather than certificate compilers. The proved Q bound below is an unconditional packing ceiling, far weaker than the desired max-fiber theorem. The proved BC statements distinguish raw support locators from the saturated codeword rays that MCA actually counts. The proved SP statement has two forms: a crude all-pairs ceiling, and the sharper structural fact that SP is automatically controlled once Q is available as a max-fiber bound. None of these statements is advertised as closing the dense frontier.

For the entropy calculations, let

$$\Xi_K(a) = \log_2 \binom{n}{a} - (a - K) \log_2 |\mathbb{B}|,$$

where $K = k$ for a list route and $K = k + 1$ for the MCA lower route. Also set

$$\varphi_\beta(g) = H_2(\rho + g) - \beta g, \quad 0 \leq g \leq 1 - \rho.$$

The number $g^*(\rho, \beta)$ is the last zero-crossing scale for φ_β : $\varphi_\beta(g) > 0$ below the envelope and $\varphi_\beta(g) < 0$ above it.

Lemma 4.1 (entropy bookkeeping for the frontier). *Let $K = k + O(1)$, $k/n \rightarrow \rho$, and $a = K + gn + O(1)$. Then*

$$\Xi_K(a) = n\varphi_\beta(g) + O(\log n + \beta).$$

Consequently, if B_n^ is a target numerator and $R(n) \geq 1$ is a ledger overhead, then*

$$R(n)2^{\Xi_K(a)} \leq B_n^*$$

holds eventually as soon as

$$n(-\varphi_\beta(g)) \geq \log_2 R(n) - \log_2 B_n^* + \omega(\log n + \beta).$$

Similarly, the prefix floor exceeds the target, $2^{\Xi_K(a)} > B_n^$, as soon as*

$$n\varphi_\beta(g) \geq \log_2 B_n^* + \omega(\log n + \beta).$$

If the residual factors and budgets are polynomial in n , and if $g^ = g^*(\rho, \beta)$ is the crossing point, then a logarithmic agreement reserve away from $K + g^*n$ suffices. More generally, for $e^{o(n)}$ residual factors, any sublinear reserve whose entropy gain dominates their logarithms suffices.*

Proof. Stirling's estimate gives, uniformly for a/n in compact subintervals of $(0, 1)$,

$$\log_2 \binom{n}{a} = nH_2(a/n) + O(\log n).$$

Since $K = k + O(1)$ and $a = K + gn + O(1)$, we have $a/n = \rho + g + O(1/n)$ and

$$(a - K) \log_2 |\mathbb{B}| = gn\beta + O(\beta).$$

This proves the displayed asymptotic for $\Xi_K(a)$. The two integer inequalities follow by exponentiating after absorbing the $O(\log n + \beta)$ term into the written $\omega(\log n + \beta)$ margin.

Finally, for fixed β , φ_β is strictly concave, $\varphi_\beta(0) = H_2(\rho) > 0$, and $\varphi_\beta(1 - \rho) = -\beta(1 - \rho) < 0$. Hence the last crossing is isolated, and on either side of it the mean-value theorem converts an agreement reserve s into an entropy reserve comparable to $|\varphi'_\beta(g^*)|s$, up to lower-order terms. This is $\Omega(s)$ for fixed β , and $\Omega(\beta s)$ in the large-base-field regime where the derivative is of order β . Thus $s = O(\log n)$ handles polynomial overheads, while an $o(n)$ reserve can be chosen to dominate any prescribed $e^{o(n)}$ overhead. If $\beta = \beta_n$ varies with n , the displayed entropy

inequalities are the safe formulation; the phrase “below the crossing” must mean below the moving zero $g^*(\rho, \beta_n)$ by enough entropy reserve to make those inequalities true. $\square$

Theorem 4.2 (identity-prefix unsafe theorem). *Let $D_n \subseteq \mathbb{B}_n \subseteq \mathbb{F}_n$, $|D_n| = n$, $k_n/n \rightarrow \rho$, $\beta_n = \log_2 |\mathbb{B}_n|$, and let B_n^* be the target numerator. Let $K_n = k_n + \eta$, where $\eta = 0$ for a list row and $\eta = 1$ for the MCA lower route, and let*

$$a_n = K_n + g_n n + O(1), \quad w_n = a_n - K_n.$$

For list rows, if

$$n\varphi_{\beta_n}(g_n) \geq \log_2 B_n^* + \omega(\log n + \beta_n),$$

then the list numerator at agreement a_n is larger than B_n^* . For MCA rows, put

$$L_n = \left\lceil \binom{n}{a_n} |\mathbb{B}_n|^{-w_n} \right\rceil, \quad q_n = |\mathbb{F}_n|.$$

If $q_n > n$ and

$$\left\lceil \frac{L_n(q_n - n)}{q_n - n + k_n(L_n - 1)} \right\rceil > B_n^*,$$

then the MCA numerator at agreement a_n is larger than B_n^* . Therefore every agreement sequence below the relevant moving crossing, in the precise sense that the displayed entropy reserve holds for $\varphi_{\beta_n}(g_n)$, is unsafe for list rows; it is unsafe for MCA rows whenever the exact simple-pole separation inequality above also holds.

Proof. By Lemma 4.1, the list hypothesis gives

$$\left\lceil \binom{n}{a_n} |\mathbb{B}_n|^{-w_n} \right\rceil > B_n^*.$$

The list conclusion is exactly Corollary 2.4(i). For MCA, the same prefix list is built in $\text{RS}[\mathbb{F}_n, D_n, k_n + 1]$; the displayed pole-separation inequality is exactly the hypothesis in Corollary 2.4(ii). That corollary gives more than B_n^* MCA-bad finite slopes at agreement a_n . Finally, if g_n stays below the last zero $g^*(\rho, \beta_n)$ by enough reserve to make the displayed entropy inequality true, then the list conclusion follows. This is deliberately stated in terms of φ_{β_n} so that fixed-base-field and growing-base-field rows use the same theorem without ambiguity. $\square$

Theorem 4.3 (Q: unconditional prefix-boundary packing bound). *Assume $1 \leq w < m \leq n$. Let $\Phi_w : \binom{D}{m} \rightarrow \mathbb{B}^w$ be the prefix map, and let $\mathcal{P}_Q(z) \subseteq \text{Fib}_w(z)$ be any pruned subfamily of a prefix fiber, for example the primitive part after quotient-pulled-back or planted members have been removed. Put*

$$t = \left\lfloor \frac{w}{2} \right\rfloor, \quad V_t(n, m) = \sum_{i=0}^t \binom{m}{i} \binom{n-m}{i}.$$

Then, for every prefix value z ,

$$|\mathcal{P}_Q(z)| \leq |\text{Fib}_w(z)| \leq \frac{\binom{n}{m}}{V_t(n, m)}.$$

Consequently the density-normalized prefix-boundary overhead satisfies

$$R_Q^{\text{pack}} := |\mathbb{B}|^w \max_z \frac{|\mathcal{P}_Q(z)|}{\binom{n}{m}} \leq \frac{|\mathbb{B}|^w}{V_t(n, m)} \leq \frac{|\mathbb{B}|^w}{\binom{m}{t} \binom{n-m}{t}}.$$

The anticode cap

$$|\mathcal{P}_Q(z)| \leq |\text{Fib}_w(z)| \leq \frac{\binom{n}{m}}{\binom{n-m+w}{w}}$$

also holds.

Proof. Use the Johnson distance $d_J(M, M') = |M \setminus M'|$ on m -subsets of D . By Proposition 3.7, two distinct members of a common prefix fiber have Johnson distance at least $w + 1$. Therefore the closed Johnson balls of radius $t = \lfloor w/2 \rfloor$ around members of $\text{Fib}_w(z)$ are pairwise disjoint: if two such balls met, the two centers would have distance at most $2t \leq w$, a contradiction.

The radius- t Johnson ball around any m -subset has size $V_t(n, m)$, because a point at distance i is obtained by deleting i elements of the center and inserting i elements from its complement. The disjoint union of all these balls is contained in the full set $\binom{D}{m}$, which has size $\binom{n}{m}$. Hence $|\text{Fib}_w(z)|V_t(n, m) \leq \binom{n}{m}$. Since $\mathcal{P}_Q(z) \subseteq \text{Fib}_w(z)$, the same bound holds for $\mathcal{P}_Q(z)$. Multiplication by $|\mathbb{B}|^w / \binom{n}{m}$ gives the normalized overhead inequality, and the final one-term bound follows from $V_t(n, m) \geq \binom{m}{t} \binom{n-m}{t}$. The anticode cap is Corollary 3.9 rewritten using

$$\frac{\binom{n}{m-w}}{\binom{m}{w}} = \frac{\binom{n}{m}}{\binom{n-m+w}{w}}.$$

□

Theorem 4.4 (BC: unconditional split-pencil dimension bound). *Fix $C = \text{RS}[\mathbb{F}, D, K]$. Let $U : D \rightarrow \mathbb{F}$, let $m \geq K$, put $\omega = n - m$, and let g_1, g_2 be a shifted weak-Popov basis of the interpolation module M_U from Proposition 3.29 with shifted degrees d_1, d_2 . Define*

$$r_i = \max\{\omega - d_i + 1, 0\} \quad (i = 1, 2).$$

The number of size- m agreement supports represented in this split-pencil profile is at most

$$\min \left\{ \binom{n}{\omega}, |\mathbb{F}|^{r_1+r_2} \right\}.$$

If the basis and the allowed coefficient pencils are defined over $\mathbb{B}$, and the cell requires $A, B \in \mathbb{B}[X]$, the second term may be replaced by $|\mathbb{B}|^{r_1+r_2}$. For an MCA line, after common-support branches have been paid first, the number of finite bad slopes carried by this profile is bounded by the same quantity.

Proof. By Proposition 3.29, every size- m agreement support in the profile is represented by

$$Ag_1 + Bg_2, \quad \deg A \leq \omega - d_1, \quad \deg B \leq \omega - d_2,$$

with the convention that no nonzero polynomial is allowed when the degree bound is negative. The two coefficient spaces therefore have dimensions r_1 and r_2 , so there are at most $|\mathbb{F}|^{r_1+r_2}$ pairs (A, B) , and each pair determines at most one locator $W = AW_1 + BW_2$. Independently, a monic D -split locator of degree ω is the locator of an ω -subset of D , so there are at most $\binom{n}{\omega}$ possible locators. This proves the first displayed bound. If the coefficient spaces are over $\mathbb{B}$, the same dimension count gives $|\mathbb{B}|^{r_1+r_2}$.

For the MCA assertion, Proposition 3.28 says that a support that is not common to the two received words supports at most one finite slope. The first-match convention assigns common-support branches before this split-pencil profile, so the remaining bad slopes are overcounted by the number of remaining locators. □

Definition 4.5 (saturated codeword rays and support census). *Let $C = \text{RS}[\mathbb{F}, D, K]$, let $U : D \rightarrow \mathbb{F}$, and put*

$$S_c(U) = \{x \in D : U(x) = c(x)\}, \quad s_c(U) = |S_c(U)|$$

for $c \in C$. The saturated ray set at agreement m is

$$\text{Ray}(U; m) = \{c \in C : s_c(U) \geq m\}.$$

Let $\omega = n - m$, and define the support-locator census

$$\begin{aligned} \text{Cen}(U; m) = \# \{ (W, N) \in \mathbb{F}[X]^2 : & W \text{ monic, } \deg W = \omega, W \mid \ell_D, \\ & WU = N \text{ on } D, W \mid N, \deg(N/W) < K \}. \end{aligned}$$

Equivalently, $W = \ell_{D \setminus T}$ for an agreement support $T \subseteq D$ of size m .

Theorem 4.6 (exact saturation identity). *With $C = \text{RS}[\mathbb{F}, D, K]$ and $\text{Cen}(U; m)$ as in Definition 4.5, for every word $U : D \rightarrow \mathbb{F}$,*

$$\text{Cen}(U; m) = \sum_{c \in C} \binom{s_c(U)}{m}.$$

Moreover, the support locators lying above a fixed codeword ray c are exactly

$$(W, N) = H (G_c, G_c c), \quad G_c = \ell_{D \setminus S_c(U)}, \quad H \mid \ell_{S_c(U)}, \quad \deg H = s_c(U) - m.$$

Consequently the fiber size of the forgetful map from support locators to saturated codeword rays is $\binom{s_c(U)}{m}$ above c .

Proof. Let (W, N) be counted by $\text{Cen}(U; m)$. Since W is monic, divides ℓ_D , and has degree $n - m$, there is a unique m -set $T \subseteq D$ such that $W = \ell_{D \setminus T}$. Since $W \mid N$ and $\deg(N/W) < K$, write $N = Wc$ for a unique codeword $c \in C$. On every $x \in T$, the value $W(x)$ is nonzero; the identity $WU = N = Wc$ on D therefore gives $U(x) = c(x)$. Thus $T \subseteq S_c(U)$.

Conversely, if $c \in C$ and $T \subseteq S_c(U)$ has $|T| = m$, then $W = \ell_{D \setminus T}$ and $N = Wc$ satisfy all defining conditions of $\text{Cen}(U; m)$. For fixed c , the number of such supports is $\binom{s_c(U)}{m}$, and summing over c gives the first identity.

For the fiber statement, write $G_c = \ell_{D \setminus S_c(U)}$. If $T \subseteq S_c(U)$, then

$$\ell_{D \setminus T} = \ell_{D \setminus S_c(U)} \ell_{S_c(U) \setminus T} = G_c H,$$

where $H = \ell_{S_c(U) \setminus T}$, hence $H \mid \ell_{S_c(U)}$ and $\deg H = s_c(U) - m$. The converse is the same construction reversed, because every monic divisor H of $\ell_{S_c(U)}$ of degree $s_c(U) - m$ is the locator of $S_c(U) \setminus T$ for a unique m -set $T \subseteq S_c(U)$. $\square$

Corollary 4.7 (raw support BC is not the MCA object). *In the setting of Theorem 4.6, if U agrees with a single codeword c on $n - d$ positions and with no other codeword on m or more positions, then*

$$|\text{Ray}(U; m)| = 1, \quad \text{Cen}(U; m) = \binom{n - d}{m}.$$

Thus an exponentially large raw support-locator family may represent one MCA ray. A split-pencil proof for MCA must therefore count saturated primitive line-rays after quotient, prefix, tangent, and common-support branches have been paid; a raw support census is a list/moment object, not the final MCA numerator.

Proof. The ray statement is the hypothesis. The support-census identity is Theorem 4.6 with one nonzero summand $\binom{s_c(U)}{m} = \binom{n - d}{m}$. The final sentence is just the same equality interpreted in MCA terms: MCA counts a bad slope once, whereas $\text{Cen}(U; m)$ counts all size- m supports below the same saturated agreement ray. $\square$

Definition 4.8 (line rays). *In the setting of Definition 4.5, for an affine line $U_Z = u + Zv$ and a set of finite slopes $E \subseteq \mathbb{F}$, define*

$$\text{LineRay}_E(u, v; m) = \{(z, c) \in E \times C : |\{x \in D : u(x) + zv(x) = c(x)\}| \geq m\}.$$

Write $s_{z,c} = |\{x \in D : u(x) + zv(x) = c(x)\}|$.

Proposition 4.9 (line-ray saturation identity). *With notation as in Definition 4.8, for every $E \subseteq \mathbb{F}$,*

$$\sum_{z \in E} \text{Cen}(U_z; m) = \sum_{(z,c) \in \text{LineRay}_E(u,v;m)} \binom{s_{z,c}}{m}.$$

Consequently

$$|\{z \in E : \exists c (z, c) \in \text{LineRay}_E(u, v; m)\}| \leq |\text{LineRay}_E(u, v; m)| \leq \sum_{z \in E} \text{Cen}(U_z; m).$$

The two inequalities are the exact deduplication losses between raw support locators, saturated codeword rays, and slope numerators.

Proof. Apply Theorem 4.6 separately to the word $U_z = u + zv$ for each $z \in E$, and then sum over z . The first inequality forgets the codeword coordinate of a line ray, and the second follows because each line ray has $s_{z,c} \geq m$, hence contributes at least one support to the right-hand side of the identity. Corollary 4.7 shows that the second inequality can be exponentially loose for a single slope and a single codeword ray, so an MCA upper ledger must either count slopes directly or control this saturation factor. $\square$

Theorem 4.10 (Q discharges the shift-pair ledger). *Let $N_w(z) = |\text{Fib}_w(z)|$, let*

$$\bar{F} = \binom{n}{m} |\mathbb{B}|^{-w},$$

and let P_e denote the full off-diagonal shift-pair stratum in Proposition 3.8, namely

$$P_e = \sum_{\substack{R \subseteq D \\ |R|=m-e}} \text{sp}_w(e; D \setminus R).$$

If Q holds in max-fiber form with constant κ ,

$$\max_z N_w(z) \leq \kappa \bar{F},$$

then necessarily $\kappa \bar{F} \geq 1$ whenever $\binom{D}{m}$ is nonempty, and

$$\sum_{e=w+1}^{\min(m, n-m)} P_e \leq \left(\kappa - \frac{1}{\bar{F}} \right) \binom{n}{m} \bar{F}.$$

Every quotient-removed or primitive shift-pair subledger satisfies the same upper bound. Hence SP is not an independent residual input in any closing theorem that already proves Q as a max-fiber theorem.

Proof. The exact second-moment identity of Proposition 3.8 says

$$\sum_z N_w(z)^2 = \binom{n}{m} + \sum_{e=w+1}^{\min(m, n-m)} P_e.$$

On the other hand,

$$\sum_z N_w(z)^2 \leq (\max_z N_w(z)) \sum_z N_w(z) \leq \kappa \bar{F} \binom{n}{m},$$

because $\sum_z N_w(z) = \binom{n}{m}$. Since at least one fiber is nonempty when $\binom{D}{m} \neq \emptyset$, the hypothesis also implies $\kappa \bar{F} \geq 1$, so the displayed right-hand side is nonnegative in the nonvacuous case. Subtracting the diagonal term $\binom{n}{m}$ gives the displayed inequality. Any primitive or quotient-removed ledger is a sub-sum of the full off-diagonal ledger, so it is bounded a fortiori. $\square$

Theorem 4.11 (SP: unconditional primitive shift-pair bound). *Use the notation of Proposition 3.27. For $w+1 \leq e \leq \min(m, n-m)$, put*

$$T_e(n, m) = \binom{n}{m-e} \binom{n-m+e}{e} \binom{n-m}{e}.$$

Then the total off-diagonal contribution at distance e satisfies

$$P_e^{\text{quot}} + P_e^{\text{prim}} \leq T_e(n, m).$$

If $D \subseteq \mathbb{B}$, then the top stratum $e = w+1$ also satisfies the sharper constant-shift bound

$$P_{w+1}^{\text{quot}} + P_{w+1}^{\text{prim}} \leq (|\mathbb{B}| - 1) \binom{n}{m-w-1} \binom{n-m+w+1}{w+1}.$$

Consequently

$$\Gamma_2 \leq \frac{1}{\bar{F}} + \frac{\sum_{e=w+1}^{\min(m, n-m)} T_e(n, m)}{\binom{n}{m} \bar{F}},$$

with the option of replacing the $e = w+1$ term by the sharper displayed top-stratum bound.

Proof. Fix e . In the exact second-moment decomposition of Proposition 3.8, an off-diagonal pair is encoded by a common part $R \subseteq D$ of size $m - e$, then by two disjoint degree- e root sets inside $D \setminus R$. There are $\binom{n}{m-e}$ choices for R . Once R is fixed, the ambient set $D \setminus R$ has size $n - m + e$; the first root set has at most $\binom{n-m+e}{e}$ choices and the second, disjoint from it, has at most $\binom{n-m}{e}$ choices. This ignores the depth equation and the quotient/primitive first-match assignment, so it is an upper bound for $P_e^{\text{quot}} + P_e^{\text{prim}}$.

When $e = w + 1$, Proposition 3.8 shows that every top-stratum shift pair has the form $A - B = c \in \mathbb{F}^\times$. If both polynomials split over $D \subseteq \mathbb{B}$, then both have coefficients in $\mathbb{B}$, hence $c \in \mathbb{B}^\times$. For each fixed R , choose A by choosing its e -element root set in $D \setminus R$, and choose $c \in \mathbb{B}^\times$; the polynomial $B = A - c$ is then determined. Requiring B to split over $D \setminus R$, to be disjoint from A , and then assigning the pair to either the quotient or primitive ledger can only reduce the count. This gives the top-stratum bound. The displayed Γ_2 inequality is Proposition 3.27 with the total quotient-plus-primitive off-diagonal contribution in every stratum replaced by the just-proved upper bound. $\square$

Proposition 4.12 (size of the proper Q bound at the active adjacent rows). *At the four active adjacent agreements, the packing estimate in Theorem 4.3 gives the following one-term upper bounds for the remaining normalized prefix overhead:*

row	w	$\log_2 R_Q^{\text{pack}}$	allowed by the bound	spare margin
KoalaBear MCA	67471		1661552.48	22.1969
KoalaBear list	67471		1661552.47	22.0109
Mersenne-31 MCA	67447		1660926.12	3.2589
Mersenne-31 list	67447		1660926.11	3.0730.

Thus the unconditional packing proof is a real theorem but is far too weak to supply the adjacent safe ledger.

Proof. For each row, $n = 2097152$, $K = k + 1 = 1048577$ on the MCA route, $K = k = 1048576$ on the list route, $w = a_+ - K$, and $\log_2 |\mathbb{B}| = 31$. The last inequality of Theorem 4.3 gives

$$\log_2 R_Q^{\text{pack}} \leq 31w - \log_2 \binom{a_+}{\lfloor w/2 \rfloor} - \log_2 \binom{n - a_+}{\lfloor w/2 \rfloor}.$$

Substituting the four active adjacent agreements 1116048, 1116047, 1116024, 1116023 gives the displayed values. For these four rows the Johnson-ball summand

$$J_i = \binom{a_+}{i} \binom{n - a_+}{i}$$

is increasing on $0 \leq i \leq t$. Indeed

$$\frac{J_i}{J_{i-1}} = \frac{(a_+ - i + 1)(n - a_+ - i + 1)}{i^2},$$

and at $i = t$ this ratio is already > 900 in all four rows, while the ratio is larger for smaller i . Hence $V_t(n, a_+) \leq (t + 1)J_t$, so replacing the one-term denominator by the full ball can lower the displayed logarithmic upper bound by less than $\log_2(t + 1) < 16$ bits. This is negligible compared with the gap between the displayed million-bit overhead and the row margins. $\square$

Theorem 4.13 (theorem-level inventory with proper bounds). *Without any dense-frontier census assumption, the following theorem-level pieces are available in this note.*

- (i) *Prefix witnesses and the simple-pole list-to-MCA conversion give the exact lower-side criteria of Proposition 3.2, Theorems 2.3 and 4.2, and Corollary 2.4.*
- (ii) *The broad safe side contains the deep-regime MCA theorem, the exact high-agreement tangent cell, MCA-from-CA below half distance, and subfield confinement, namely Theorems 2.10, 2.13 and 2.14 and Proposition 2.11.*
- (iii) *The residual cells have unconditional theorem-level pieces: Q has the Johnson packing and anticode bounds of Theorem 4.3; raw split-pencil support counts have the dimension bound of Theorem 4.4; MCA support multiplicities are exactly saturated by Theorem 4.6,*

Corollary 4.7, and Proposition 4.9; and SP is downstream from Q by Theorem 4.10, with the all-pairs ceiling of Theorem 4.11 available as an audit bound.

- (iv) The stronger dense-frontier conclusion is not proved by these bounds. In particular, Proposition 4.12 shows that the proved Q packing bound alone leaves overhead far beyond the active adjacent margins, and Corollary 4.7 shows that the MCA split-pencil target must be saturated before it can be an MCA numerator.

Proof. Parts (i) and (ii) are the cited theorems. Part (iii) is exactly Theorems 4.3, 4.4, 4.6, 4.10 and 4.11, Corollary 4.7, and Proposition 4.9. Part (iv) follows from Proposition 4.12: an upper estimate that permits more than $2^{1.66 \cdot 10^6}$ normalized prefix overhead cannot imply a ledger with only $2^{22.2}$ or less spare overhead. The saturation identity also changes the MCA split-pencil target from raw supports to primitive line-rays. Therefore the adjacent safe inequality requires stronger row-wise arguments than the unconditional replacement bounds proved here. $\square$

Theorem 4.14 (deployed-row theorem-level status). *For the four deployed rows and budgets displayed in Remark 2.1, the following unconditional conclusions follow from the banked integer arithmetic and the proper bounds in this note.*

- (i) The active agreements a_0 in the row table are unsafe.
- (ii) The identity-prefix lower route stops at $a_0 + 1$; this is not an upper bound.
- (iii) The broad safe anchors listed in Proposition 2.7 start strictly above $a_0 + 1$, so they do not prove the adjacent safe inequality.
- (iv) The unconditional Q packing bound, raw split-pencil dimension bound, saturation identity, and SP-from-Q theorem of Theorems 4.3, 4.4, 4.6 and 4.10 are proved, but they do not by themselves imply $B(a_0 + 1) \leq B^*$ for any of the four adjacent rows.

Proof. Part (i) and the lower-route stopping statement in part (ii) are Proposition 2.5. Part (iii) is Proposition 2.7. Part (iv) follows because the first residual replacement bound already has the gap quantified in Proposition 4.12; the saturation theorem changes the object to be counted but does not supply the needed saturated primitive line-ray bound; and Theorem 4.10 only discharges SP after a sufficiently sharp Q max-fiber theorem has been proved. Hence none of the proved statements supplies the missing inequality $B(a_0 + 1) \leq B^*$. $\square$

Theorem 4.15 (adjacent criterion from an explicit upper bound). *Fix one of the four deployed rows in the table, with its relevant numerator $B(a)$ from Definition 1.3. Suppose the unsafe inequality at a_0 is the banked identity-prefix theorem, and suppose a proved upper bound gives*

$$B(a_0 + 1) \leq U(a_0 + 1) \leq B^*.$$

For an MCA row, U may be obtained by a first-match, deduplicated sum of tangent, quotient, extension, prefix-boundary, split-pencil-interior, primitive shift-pair, and any explicitly named residual cells. For a list row, U is the corresponding exact list upper bound. Then the first safe integer agreement is $a_0 + 1$, and $\delta_C^(\epsilon^*)$ is pinned to one integer step under the closed-ball convention for that row object.*

Proof. By the unsafe theorem, $B(a_0) > B^*$, so Lemma 1.4 gives error $> \epsilon^*$ at agreement a_0 . The proved upper bound gives $B(a_0 + 1) \leq U(a_0 + 1) \leq B^*$. Applying Lemma 1.4 again gives error at most ϵ^* at agreement $a_0 + 1$. Monotonicity makes every agreement threshold $a \leq a_0$ unsafe once a_0 is unsafe, so $a_0 + 1$ is the first safe integer agreement. This is precisely the adjacent staircase condition of Definition 1.3. In the MCA case, Lemma 1.6 is the generic mechanism by which a first-match cell ledger proves the required upper inequality. $\square$

Remark 4.16 (endpoint convention). If Theorem 4.15 is instantiated with a proved upper bound at $a_0 + 1$, then the largest safe closed-grid radius is $(n - a_0 - 1)/n$. The first proved unsafe grid radius is $(n - a_0)/n$, and monotonicity makes every larger closed-grid radius unsafe. For MCA this is Lemma 1.2; for list rows it is the immediate fact that a codeword list at agreement a is still valid at every lower agreement threshold. If one records $\delta_C^*(\epsilon^*)$ as a real supremum over closed integer balls, the threshold lies at the upper edge of this unit interval and the convention must be printed with the proof.

5. AUDIT CORRECTIONS AND EXPANDED GAP LEDGER

The preceding sections now leave three kinds of statements, and the distinction is part of the proof.

Proposition 5.1 (closed audit of what is and is not proved). *In this expanded version:*

- (i) *The lower-side identity-prefix and simple-pole implications are proved theorems under their stated field-size hypotheses.*
- (ii) *The broad safe anchors and the high-agreement tangent/common-line cells are proved only in their displayed agreement ranges.*
- (iii) *The Q, BC, and SP replacements in Theorems 4.3, 4.4, 4.6, 4.10 and 4.11 and Proposition 4.9 are unconditional statements, but their numerical strength is insufficient for the four active adjacent agreements.*
- (iv) *No statement in this note proves $B(a_0 + 1) \leq B^*$ for any deployed adjacent row.*

Proof. Part (i) is Proposition 3.2, Theorems 2.3 and 4.2, and Corollary 2.4. Part (ii) is Theorems 2.10, 2.13 and 2.14 and Propositions 2.7 and 2.11. Part (iii) follows from the cited residual-cell theorems and from Proposition 4.12. Part (iv) is then immediate from Theorem 4.14: the only theorem in this note that would turn the banked unsafe edge into a closed adjacent threshold is Theorem 4.15, and its upper-bound hypothesis is not supplied by the proved replacement bounds. $\square$

Problem 5.2 (row-sharp multilevel Q target). *For each active adjacent row, after quotient-pulled-back, planted, tangent, and explicitly paid lower-dimensional cells have been assigned by first match, prove a max-fiber bound of the form*

$$\max_z |\mathcal{P}_Q(z)| \leq R_Q^{\text{row}} \binom{n}{a_+} |\mathbb{B}|^{-(a_+ - K)}$$

with the exact row-specific normalized overhead R_Q^{row} fitting the remaining integer budget. A proof through moments must satisfy the finite inequality

$$\log_2 \Gamma_r + (a_+ - K) \log_2 |\mathbb{B}| \leq r \Delta_Q$$

with the actual allocated bit margin Δ_Q , not just an $e^{o(n)}$ asymptotic loss.

Problem 5.3 (saturated primitive split-pencil line-ray target). *For each active MCA adjacent row, after quotient, boundary-Q, common-support, tangent, and extension branches have been paid by first match, bound the number of residual bad slopes through saturated primitive line rays:*

$$|\{z : \exists c (z, c) \in \text{LineRay}_E(u, v; a_+) \text{ and } z \text{ is residual-bad}\}|.$$

By Proposition 4.9, a raw support-locator census is only an upper audit object,

$$\sum_{z \in E} \text{Cen}(u + zv; a_+) = \sum_{(z, c) \in \text{LineRay}_E(u, v; a_+)} \binom{s_{z, c}}{a_+},$$

and can be exponentially larger than the slope numerator. The required theorem must therefore control either the distinct slope set directly or the saturated line-ray set with a proved multiplicity loss small enough for the exact row margin.

Remark 5.4 (finite arithmetic artifacts). The exact integer tables in this note can be checked independently from the displayed formulas using integer arithmetic only: $B^* = \lfloor p^s 2^{-\lambda} \rfloor$, $L(a) = \lceil \binom{n}{a} p^{-(a-K)} \rceil$, the simple-pole value $M(a) = \lceil L(a)(q - n)/(q - n + k(L(a) - 1)) \rceil$, the extension headroom in Proposition 2.8, and the logarithmic Q-packing estimate in Proposition 4.12. These checks are arithmetic audits; they do not replace the missing upper-ledger theorems in Problems 5.2 and 5.3.

6. REMAINING FINITE UPPER-LEDGER TARGET

The proper proofs above remove the fake certificate-theorem closure. They prove real replacement statements, but they do not settle the dense frontier. A finite adjacent proof must improve the upper bound $U(a_0 + 1)$ in Theorem 4.15 until it fits inside the exact margins 22.1969, 22.0109, 3.2589, 3.0730 bits. After Theorem 4.10, the independent active MCA targets are multilevel Q and saturated primitive BC; SP remains an exact audit ledger and follows once Q is strong enough.

Prefix-boundary ledger. The prefix-boundary row must bound the actual maximum fiber, not the raw null fiber. Proposition 3.14 rules out the mode-at-null shortcut, and Theorem 3.11 gives the exact moment inequality required for a finite proof:

$$\log_2 \Gamma_r + w \log_2 |\mathbb{B}| \leq r\Delta.$$

The new unconditional packing theorem Theorem 4.3 is valid but, by Proposition 4.12, is far too weak at the deployed adjacent rows.

Split-pencil interior ledger. The split-pencil proof must count saturated primitive line-rays, not raw support locators. The exact saturation identity Theorem 4.6 explains the difference, and Corollary 4.7 shows why the raw count can be exponentially larger than the MCA numerator. The remaining proof must improve the dimension bound of Theorem 4.4 by using the actual split condition, quotient removal, base-field normalization, saturation, and row-specific chart equations. The base-field floor in Theorem 3.37 shows that challenge-field normalization alone is false.

Primitive shift-pair ledger. The primitive pair ledger should remain in the paper as the exact L^2 identity of Proposition 3.27. It is no longer an independent active assumption in the MCA closure: Theorem 4.10 proves that a max-fiber Q theorem bounds the whole SP ledger. The unconditional count in Theorem 4.11 is only an all-pairs audit ceiling.

Rung audit. The exact divisor-lattice audit at $a_0 + 1$ should cover all quotient scales. If a periodic rung exceeds the budget, the adjacent candidate fails for a paid reason and the edge moves. If every rung is below budget with printed margin, the remaining proof target is genuinely primitive.

7. CONCLUSION

This version contains no Q, BC, or SP certificate theorem. The unsafe side is exact and the broad safe anchors are unconditional in their stated ranges. The former SP input is now a theorem downstream from Q; the raw BC support count is replaced by exact saturation and line-ray identities; and the unconditional Q/BC/SP ceilings that remain are explicitly proved but too weak for the deployed adjacent rows. The full entropy-subfield equality and the adjacent safe inequalities are therefore not asserted. A closed adjacent theorem still requires a row-sharp multilevel Q max-fiber bound and a saturated primitive split-pencil line-ray bound satisfying Theorem 4.15.

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